TRIBHUVAN UNIVERSITY
INSTITUTE OF SCIENCE AND TECHNOLOGY

A Project on

Job Recommendation System

Submitted to:

Department of Statistics and Computer Science
Patan Multiple Campus

In partial fulfilment of the requirements of the Bachelor's Degree in Computer Science and Information Technology

Under the Supervision of
Mr. Bikash Balami

Submitted by :

Swastika Dangol (79010132)
Ankita Gautam (79012304)
Rojina Shrestha(79010091)

July, 2026

CERTIFICATION OF DECLARATION

We hereby declare that the project entitled "Job Recommendation System" submitted by Swastika Dangol, Ankita Gautam, and Rojina Shrestha is our original work carried out under the supervision of Mr. Bikash Balami in partial fulfilment of the requirements for the degree of Bachelor of Science in Computer Science and Information Technology (BSc CSIT).

We further declare that this project has not been submitted, either in whole or in part, to any other university or institution for the award of any degree or academic qualification. All sources of information and materials used in this project have been properly acknowledged and cited in accordance with academic standards.

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SUPERVISOR’S RECOMMENDATION

This is to certify that the project entitled "Job Recommendation System" has been carried out by Swastika Dangol, Ankita Gautam, and Rojina Shrestha under my supervision as partial fulfilment of the requirements for the degree of Bachelor of Science in Computer Science and Information Technology (BSc CSIT).

To the best of my knowledge, this project is an original work and has been completed satisfactorily. I believe that it fulfils the academic requirements of the Department of Computer Science, Patan Multiple Campus, Tribhuvan University, and I recommend it for evaluation and acceptance.

…………………………….

Bikash Balami

Supervisor

Patan Multiple Campus

LETTER OF APPROVAL

This is to certify that the project entitled "Job Recommendation System" submitted by Swastika Dangol, Ankita Gautam, and Rojina Shrestha in partial fulfillment of the requirements for the degree of Bachelor of Science in Computer Science and Information Technology (BSc CSIT) has been examined and approved by the Department of Computer Science, Patan Multiple Campus, Tribhuvan University.

The project has been carried out under the supervision of Mr. Bikash Balami and is considered satisfactory in terms of scope, quality, and academic contribution.

…………………………

Bikash Balami

Supervisor

Patan Multiple Campus

…………………………

External Examiner

Tribhuvan University

…………………………

Internal Examiner

Patan Multiple Campus

ACKNOWLEDGEMENT

We would like to express our sincere gratitude to everyone who contributed to the successful completion of our project entitled "Job Recommendation System."

First and foremost, we would like to express our heartfelt gratitude to our project supervisor, Mr. Bikash Balami, for his continuous guidance, valuable suggestions, constructive feedback, and encouragement throughout the development of this project. His knowledge, support, and motivation were invaluable in the successful completion of this work.

We are also sincerely grateful to the Department of Computer Science and all the faculty members of Patan Multiple Campus, Tribhuvan University, for providing us with the opportunity, academic guidance, and necessary resources to carry out this project successfully. Their support and encouragement throughout our undergraduate studies have been instrumental in enhancing our knowledge and skills.

Finally, we would like to express our deepest gratitude to our parents, family members, friends, and everyone who directly or indirectly supported and encouraged us throughout the completion of this project. Their constant encouragement, patience, and moral support have been a great source of motivation in achieving this milestone.

ABSTRACT

The Job Recommendation System is a web-based recruitment platform developed to simplify and improve the job search and recruitment process for job seekers, employers, and administrators. The system enables job seekers to create and manage profiles, upload and manage resumes, generate professional CVs in PDF format, browse job vacancies, receive personalized job recommendations, and apply for suitable positions. Employers can create and manage job postings, use the Google Gemini AI service to automatically generate job descriptions, responsibilities, and required skills, and review applications submitted by candidates. Administrators are responsible for managing users and monitoring the overall operation of the system.

The recommendation module employs a hybrid content-based recommendation approach that integrates skill normalization, TF-IDF feature weighting, Cosine Similarity, weighted scoring, and job ranking to recommend jobs based on the similarity between candidate profiles and job requirements. The system is developed using React.js for the frontend, ASP.NET Core Web API for the backend, Entity Framework Core for database interaction, and Microsoft SQL Server for data storage. Secure authentication and authorization are implemented using JSON Web Tokens (JWT).

The developed system enhances the recruitment process by reducing manual effort in candidate-job matching, improving the relevance of job recommendations, assisting employers in preparing job postings through AI-generated content, and enabling users to generate professional CVs automatically. The implementation and testing results demonstrate that the proposed system satisfies the specified functional requirements while providing a secure, reliable, and user-friendly recruitment platform.

Keywords: Job Recommendation System, Content-Based Recommendation, TF-IDF, Cosine Similarity, Google Gemini AI, CV Generation, React.js, ASP.NET Core Web API.

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LIST OF ABBREVIATIONS

AI	:	Artificial Intelligence
API	:	Application Programming Interface
CV	:	Curriculum Vitae
EF Core	:	Entity Framework Core
HTML	:	Hypertext Markup Language
IDF	:	Inverse Document Frequency
JWT	:	JSON Web Token
JSON	:	JavaScript Object Notation
PDF	:	Portable Document Format
REST	:	Representational State Transfer
SQL	:	Structured Query Language
TF	:	Term Frequency
TF-IDF	:	Term Frequency – Inverse Document Frequency
UI	:	User Interface
URL	:	Uniform Resource Locator
HTTP	:	HyperText Transfer Protocol
HTTPS	:	HyperText Transfer Protocol Secure
ORM	:	Object Relational Mapping

Chapter 1: Introduction

1.1 Introduction

As the number of online job portals is increasing rapidly, it is becoming challenging for students as well as job seekers to find appropriate internships and jobs according to their skills and interests. The existing systems mostly provide general search results, but they do not recommend jobs in a personalized manner. This results in a time-consuming process for the users, especially fresh graduates and student who want to search for the most suitable internships.

A Job Recommendation System is a web-based application that recommends suitable internships and jobs to the users based on their individual profiles. The system compares the users information, such as skills, education and interests, with the job requirements provided by the companies and recommends the suitable and appropriate internships and jobs to the users.

Users can add their skills, education, experience in their profile information, based on which the system recommend the jobs and internship to them that match their profile. Whereas the job provider company can post vacancies of internship or job on the system which clearly explain the required skills and qualifications for the position. The system compares the requirement posted by company and the information of users and suggests the suitable internships and jobs to the users.

The system will be developed using a web-based platform that can be accessed by users from anywhere. The system will have a user-friendly interface that enables students and job seekers to create their own profile and receive personalized recommendations. The system will use a content-based recommendation approach to match user profiles with job and internships description. The system will use basic technique like content-based to identify important skills and cosine similarity to identify the best match. In addition, users will be able to filter results by type, location.

Therefore, this project aims to develop a smart and efficient recommendation system that makes the job and internship searching process easier, faster and more accurate.

1.2 Problem Statement

Before the introduction of online platforms for jobs and internships, the process of searching for jobs and internships was difficult and time-consuming. Students and job seekers had to manually visit different offices or rely on notice boards and personal contacts to find the required and suitable jobs and internships. The process was confusing, stressful and inefficient, especially for fresh graduates and students who were not always aware of which positions matched their skills, education and career goals. many opportunities were missed simply due to lack of information or access. Similarly, the employer also faced difficulties in searching for the suitable candidates for the available positions. Overall, both students and employers struggled due to lack of a structed, organized and accessible platform, resulting in wasted time, effort and missed opportunities for suitable matches.

1.3 Objectives

The main objectives of Job Recommendation System are:

- To develop a secure and user-friendly recruitment platform for job seekers, employers, and administrators.
- To implement a hybrid content-based recommendation approach using TF-IDF, Cosine Similarity, and weighted scoring to recommend suitable jobs based on candidate skills and job requirements
- To assist employer in efficiently identifying appropriate candidates, thereby reducing time and effort in the recruitment process
- To integrate Google Gemini AI for automatically generating job descriptions, responsibilities, and required skills from a given job title, reducing employers' manual effort.

1.4 Scope and Limitation

1.4.1 Scope

The proposed Job Recommendation System is a web-based application designed to meet the demands of job seekers and employers. The system is designed to allow job seekers to create and manage their profiles, upload resumes, search for jobs, get recommended employment jobs and apply for suitable positions. Employers can post job vacancies, manage applications, review candidate profiles, and evaluate match scores to support recruitment decisions.

The system also provides secure user authentication, role-based access control, and an intelligent recommendation engine based on Cosine Similarity to recommend jobs according to a candidate's skills. As a web application, the system can be accessed through any modern web browser with an internet connection.

1.4.2 Limitations

The proposed system has the following limitations:

- The recommendation results depend on the accuracy and completeness of the skills and profile information provided by job seekers.
- The recommendation algorithm primarily considers skill similarity and optional filters such as job type and work mode; it does not currently incorporate machine learning or behavioural analysis.
- The system is designed as a web application and requires an active internet connection for access.

1.5 Development Methodology

The Job Recommendation System was developed using the Waterfall Software Development Life Cycle (SDLC) model. The Waterfall model is a sequential software development approach in which each phase is completed before proceeding to the next phase. It provides a structured framework for software development and is suitable for projects with clearly defined requirements **[9]**.

The development process of the proposed system consists of the following phases:

1. **Requirement Analysis**
   During this phase, the functional and non-functional requirements of the system were gathered by studying the existing recruitment process and identifying the needs of job seekers, employers, and administrators. The project requirements were analyzed to determine the system features, constraints, and objectives.
2. **System Design**
   During this phase, the functional and non-functional requirements of the system were gathered by studying the existing recruitment process and identifying the

needs of job seekers, employers, and administrators. The project requirements were analyzed to determine the system features, constraints, and objectives.

3. **Implementation**

   The designed system was implemented using **React.js** for the frontend, **ASP.NET Core Web API** for the backend, **Entity Framework Core** for database interaction, and **SQL Server** for data storage. The recommendation module was implemented using the **Cosine Similarity** algorithm to calculate candidate-job match scores.

4. **Testing**

   The completed system was tested using unit testing, integration testing, and system testing to verify that all modules function correctly and satisfy the specified requirements.

5. **Deployment and Maintenance**

   After successful testing, the application was deployed for use. Continuous maintenance is performed to fix bugs, improve performance, and introduce new features based on user feedback and changing recruitment requirements.

1.6 Report Organization

This report has been prepared following the guidelines prescribed by Tribhuvan University. It is organized into preliminary pages and six main chapters. The preliminary pages include the Cover Page, Title Page, Recommendation, Approval Sheet, Certificate, Acknowledgement, Abstract, Table of Contents, List of Figures, List of Tables, and List of Abbreviations.

The main report is divided into the following six chapters:

Chapter 1: Introduction

This chapter introduces the Smart Job Recommendation System and provides an overview of the project. It presents the background of the study, problem statement, objectives, scope and limitations, development methodology, and the organization of the report.

Chapter 2: Background Study and Literature Review

This chapter reviews the concepts and technologies related to online job portals, recommendation systems, and machine learning techniques used in job recommendations. It also analyzes existing job portal systems, identifies their limitations, and highlights the need for the proposed system.

Chapter 3 : System Analysis and Design

This chapter presents the functional and non-functional requirements of the system, feasibility analysis, use case analysis, and system architecture. It also includes UML diagrams, Data Flow Diagrams (DFDs), Entity Relationship (ER) Diagram, database design, and user interface design.

Chapter 4: System Implementation

This chapter describes the implementation of the Smart Job Recommendation System using ASP.NET Core Web API, React.js, SQL Server, and Entity Framework Core. It explains the implementation of different system modules, recommendation logic based on skill matching, authentication and authorization, and key code snippets.

Chapter 5: Testing

This chapter presents the testing process carried out to verify the correctness and functionality of the system. It includes unit testing, integration testing, functional testing, test cases, system screenshots, recommendation results, and discussion of the overall system performance.

Chapter 6 : Conclusion and Future Recommendations

This chapter summarizes the overall project, discusses the achievements and limitations of the developed system, and provides recommendations for future enhancements, such as incorporating advanced machine learning algorithms, personalized recommendation techniques, resume parsing, and interview preparation features.

Chapter 2 : Background Study and Literature Review

2.1 Background Study

The advancement of information technology has significantly transformed the recruitment process by enabling organizations to conduct hiring activities through online platforms. Web-based recruitment systems allow employers to publish job vacancies, receive applications, and manage candidate information efficiently, while job seekers can search and apply for jobs regardless of their geographical location. Compared to traditional recruitment methods, online recruitment systems improve accessibility, reduce paperwork, and simplify communication between employers and applicants [1].

Despite these advantages, many existing recruitment platforms still rely on keyword-based searching and manual resume screening, making it difficult to identify the most suitable candidates. Likewise, job seekers often spend considerable time searching through numerous job postings that may not match their qualifications, skills, or experience. These limitations have increased the need for intelligent recommendation systems that can automatically recommend relevant job opportunities and improve the efficiency of the recruitment process [1][2].

The proposed Job Recommendation System addresses these challenges by implementing a content-based recommendation approach that compares the skills of job seekers with the required skills of available job postings. The system calculates the similarity between candidate profiles and job requirements using the Cosine Similarity algorithm, allowing users to receive personalized job recommendations while assisting employers in identifying suitable candidates more efficiently [2][3].

2.2 Literature Review

Recommendation systems have become an important research area due to their ability to provide personalized suggestions based on user preferences and item characteristics. Different recommendation approaches, including content-based, collaborative, and hybrid filtering, have been proposed to improve recommendation accuracy. Among these approaches, content-based recommendation is considered suitable for applications where user profiles and item attributes are available for comparison [2].

Similarity measurement techniques play an important role in recommendation systems. Various studies have shown that Cosine Similarity is an effective method for measuring the similarity between two vectors and is widely used in information retrieval and recommendation applications. Its computational efficiency and ability to compare textual data make it appropriate for skill-based job recommendation systems [2] [3].

Several studies have also highlighted the growing importance of intelligent recruitment systems in improving the hiring process. Traditional recruitment platforms mainly rely on keyword-based searching and manual resume screening, which often results in inefficient candidate-job matching. Integrating recommendation techniques into recruitment platforms helps reduce manual effort and provides more relevant job recommendations to applicants [1][8].

Although existing studies have proposed various recommendation approaches, many systems do not adequately consider the similarity between candidate skills and job requirements. The proposed Job Recommendation System addresses this limitation by implementing a content-based recommendation approach using the Cosine Similarity algorithm to recommend jobs based on the similarity between candidate skills and required job skills.

Chapter 3 : System Analysis

3.1 System Analysis

System analysis can be defined as the process of examining the requirements of a system that is intended to be designed and developed. It aims at discovering the needs of the users, analyzing the problems of the existing system, and establishing the requirements for the proposed system. The main purpose of system analysis is to create a system that serves the needs of the users properly.

Job recommendation system is designed to meet the needs of job seekers, employers, and administrators. The system will be efficient since it will offer services that include managing the profiles of the recruiters, managing the job ads, managing applications, recommending jobs based on the applicant's skills, managing the recommended applicants, and managing the log-in of the users. The system will be able to recommend jobs based on the skills of the applicants. It will also enable faster application processes because it will contain all the required information of the job seekers and the employers.

3.1.1 Requirement Analysis

Requirement Analysis is the study of what the system should do and how it should behave. This is one of the significant steps in software development because it enables developers and other skate holders to agree on the system's requirements before it is developed. Job Recommendation System helps students and job seekers find the suitable internships and jobs based on their skills, education and experiences.

3.1.1.1 Functional Requirements

The functional requirements describe the services provided by the system.

Job Seeker

- Register and log in to the system.
- Manage personal profile, education, experience, and skills.
- Upload and update resume.
- Browse and search available jobs.
- Receive personalized job recommendations.

- Apply for jobs.
- Generate a professional CV in PDF format.

Employer

- Register and log in.
- Create and manage company profile.
- Post internship and job vacancies.
- View applicants per jobs posted.
- View candidate match scores.
- Manage applications.
- Automatically generate responsibilities and required skills based on the job title.
- View reports.

Administrator

- Manage users.
- Approve or reject job postings.
- Monitor system activities.
- Manage system records.

3.1.1.2 Non-Functional Requirements

The proposed system should satisfy the following non-functional requirements:

a. **Performance:** The system provides a responsive user experience by efficiently processing user requests such as user authentication, profile management, job searching, job applications, and recommendation generation. The recommendation algorithm analyzes user skills, education, and experience to generate suitable job and internship recommendations within an acceptable time.

b. **Usability:** The system provides a simple, attractive, and user-friendly interface that allows job seekers, employers, and administrators to easily navigate through different functionalities. Users can perform tasks such as registration, profile creation, job posting, searching opportunities, applying for jobs, and viewing recommendations without complexity.

c. **Reliability:** The system provides consistent and accurate job and internship recommendations based on user profile information. It ensures stable operation during user activities such as managing profiles, posting jobs, applying for opportunities, and retrieving recommendations.

d. **Security:** The system implements authentication and authorization mechanisms to protect user accounts and information. Role-based access control ensures that job seekers, employers, and administrators can access only the functionalities assigned to their respective roles.

e. **Maintainability:** The system is developed using a structured and modular approach with organized code and database design. This makes the system easier to debug, maintain, and enhance with additional features in the future.

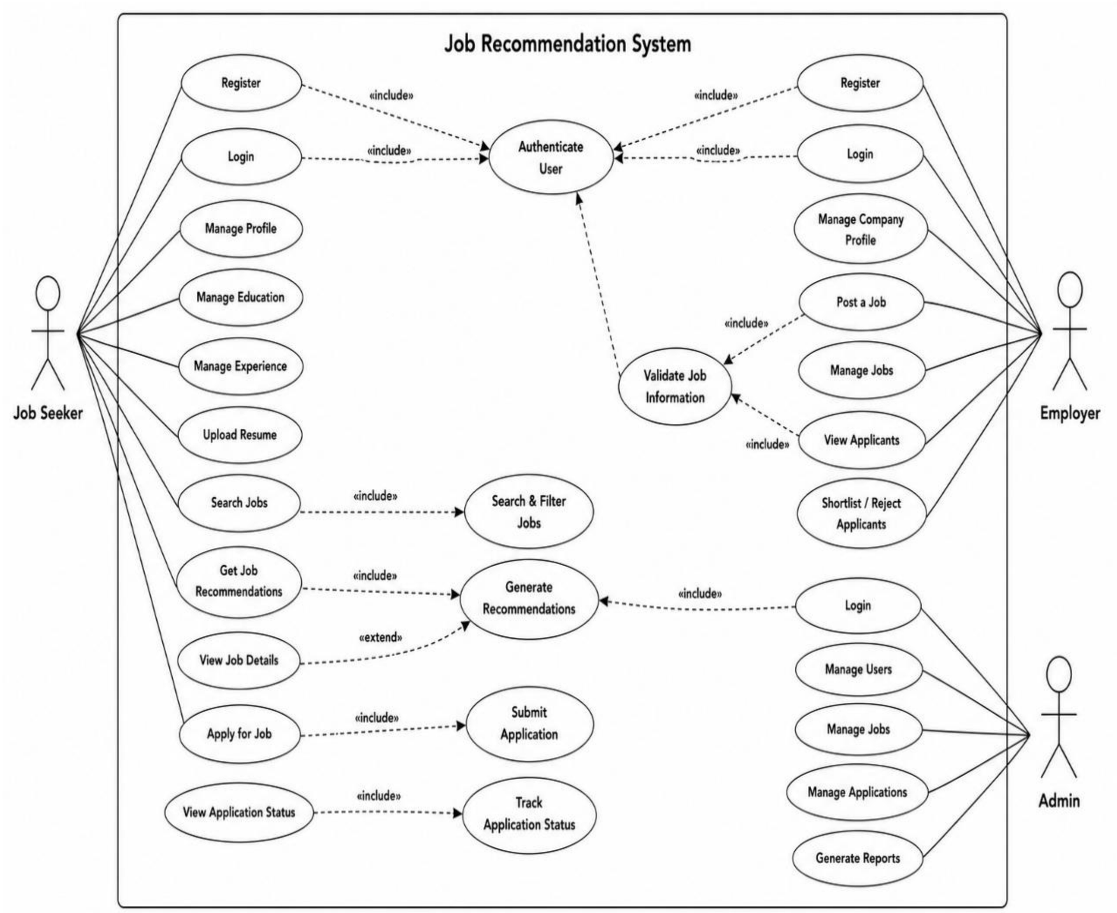

Figure 1: Use Case Diagram of Job Recommendation System

3.1.2 Feasibility Study

A feasibility study evaluates whether the proposed system can be successfully developed and implemented. It helps determine the practicality of the project from different perspectives.

3.1.2.1 Technical Feasibility

The proposed system is technically feasible because it is developed using modern and widely adopted technologies such as React.js, ASP.NET Core Web API, SQL Server, and Entity Framework Core. These technologies provide the necessary tools to develop a secure, scalable, and efficient web application.

3.1.2.2 Economic Feasibility

The project is economically feasible because it utilizes open-source frameworks and development tools. Most of the software required for development, including Visual Studio Community, Visual Studio Code, .NET, and React, is freely available, minimizing development costs.

3.1.2.3 Operational Feasibility

The proposed system is user-friendly and designed for both technical and non-technical users. The graphical user interface is simple and intuitive, allowing employers, job seekers, and administrators to perform their tasks without requiring specialized technical knowledge.

3.1.2.4 Schedule Feasibility

Schedule feasibility assesses whether a project can be completed within the planned timeframe. The Job Recommendation System was completed according to the proposed schedule. All major phases, including requirement analysis, system design, implementation, testing, and documentation, were carried out within the allocated time.

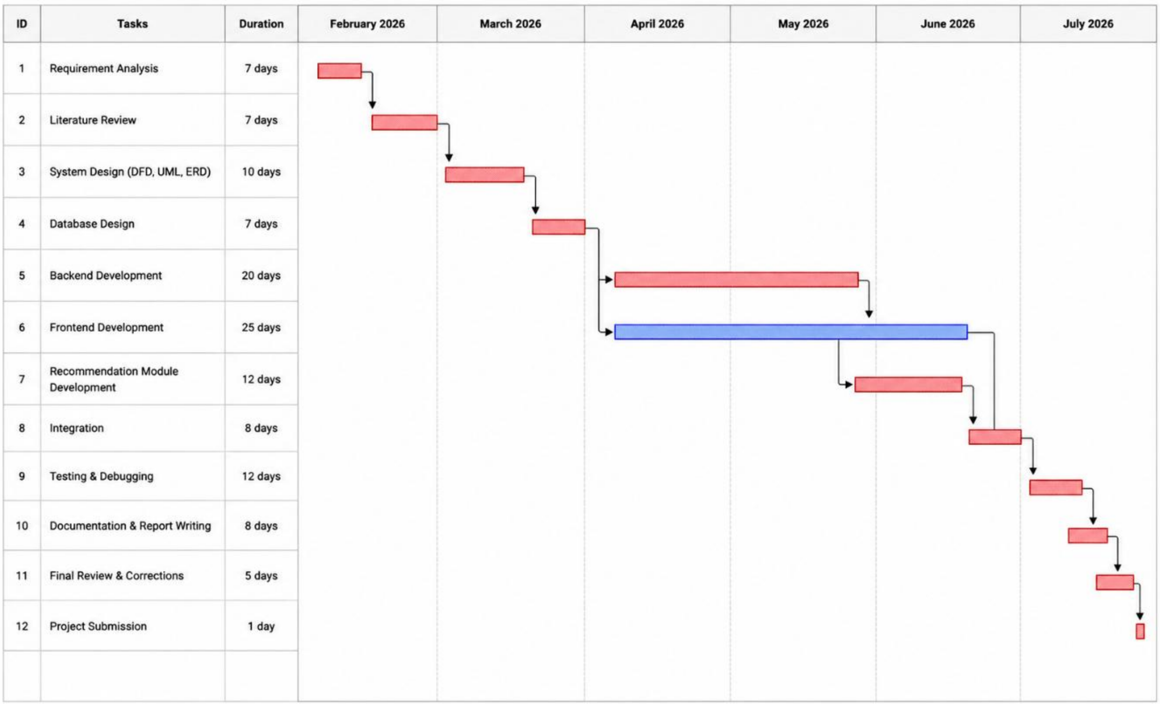

Figure 2: Gantt Chart

3.1.3 System Flowchart Diagram

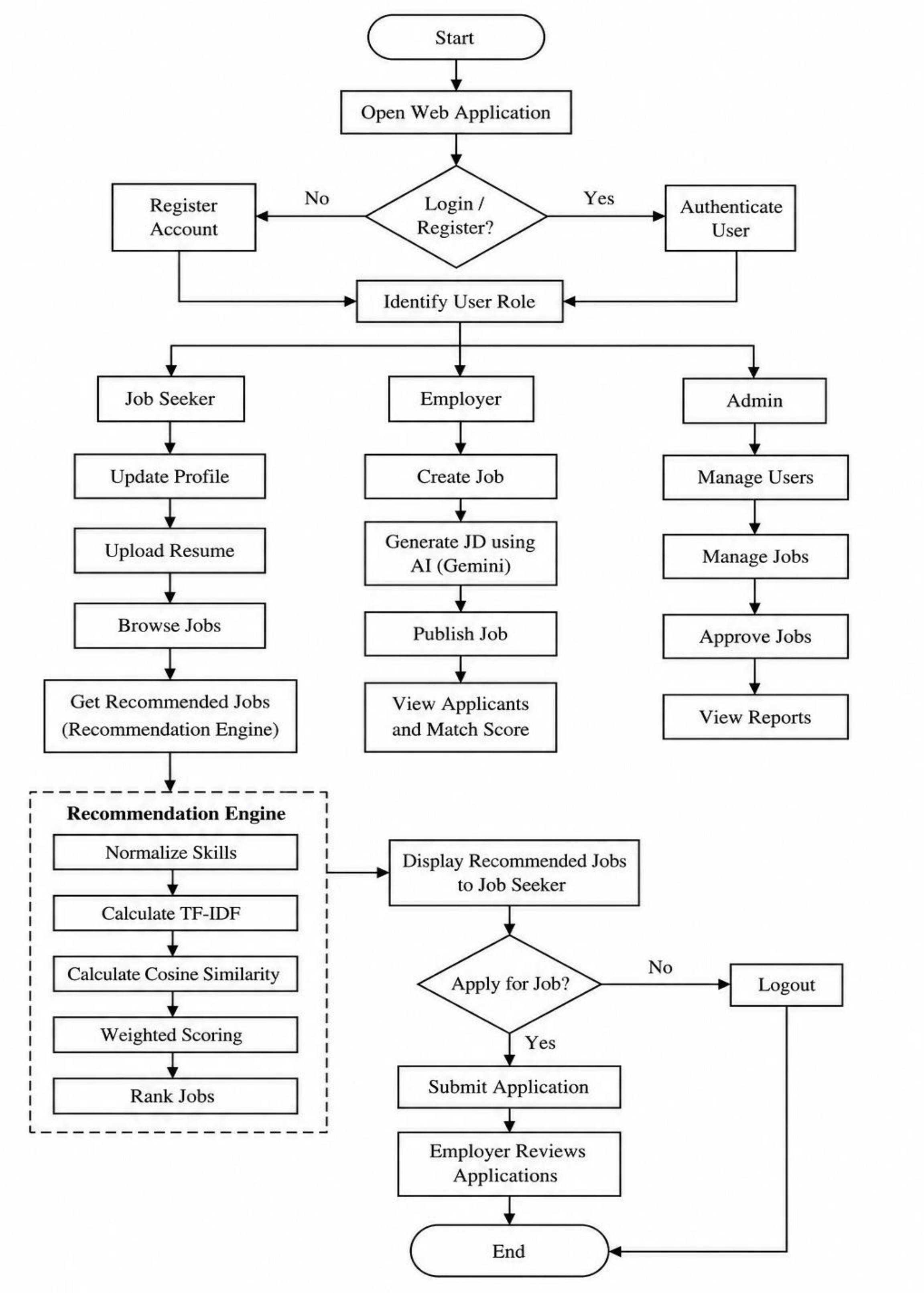

Figure 3: System Flowchart Design

3.1.4 Analysis

System analysis was performed to understand the functional requirements, data flow, and interactions between different components of the Smart Internship and Job Recommendation System. The system was analyzed using Entity Relationship (ER) Diagram for database modelling and Data Flow Diagram (DFD) for process modelling.

ER Diagram

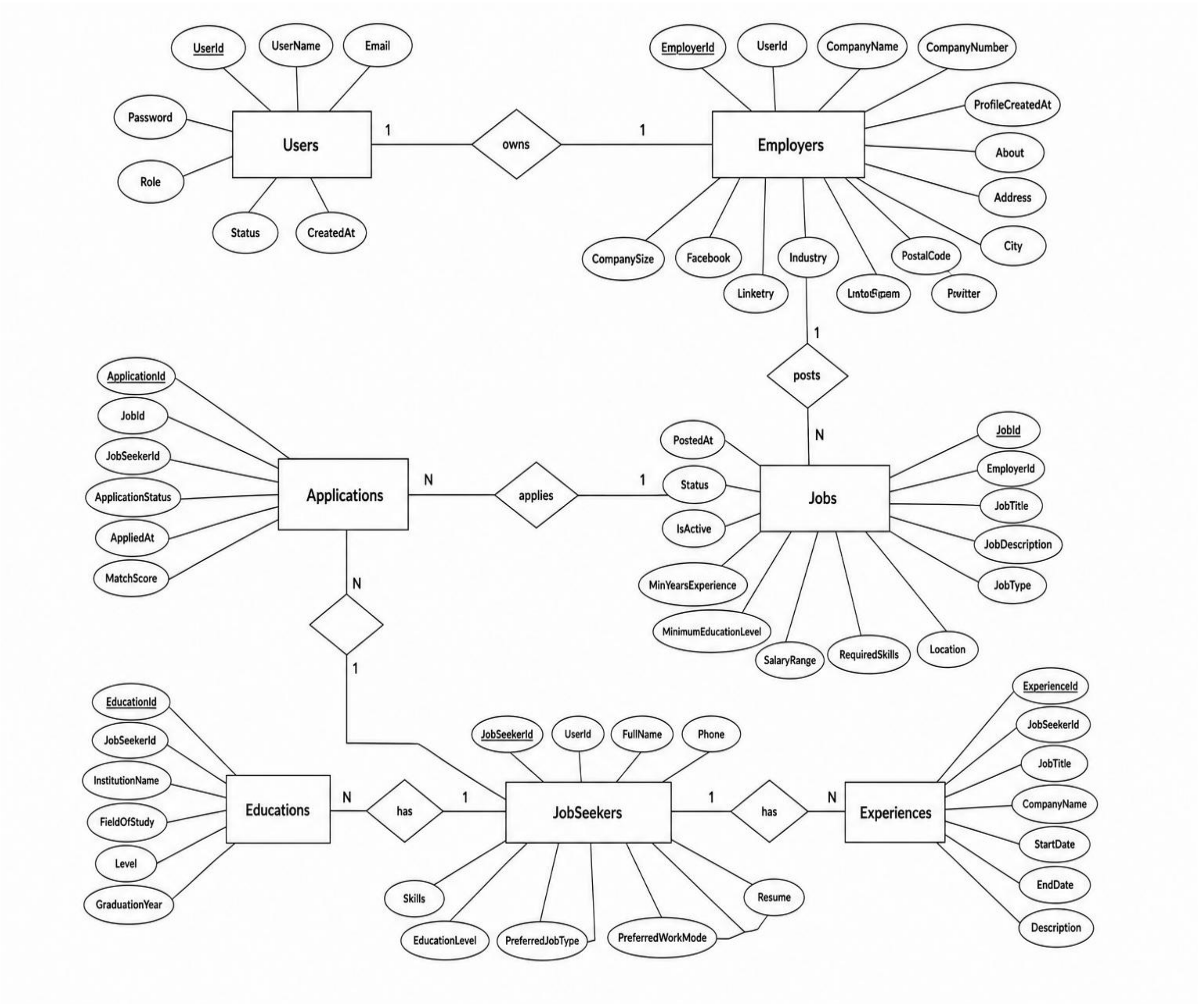

Figure 4: ER Diagram

The ER diagram represents the database structure of the Job Recommendation System. The Users entity manages authentication and connects with Job Seekers and Employers. Employers can post jobs, while Job Seekers maintain their profiles with skills, education, experience, and resume details. The Applications entity manages the relationship between Job Seekers and Jobs by storing application details and status. The stored profile and job information are used by the recommendation system to match candidates with suitable job opportunities based on their skills and requirements.

DFD Level 0

The context-level data flow diagram represents the overall view of the Job Recommendation System. It shows the interaction between external entities and the system.

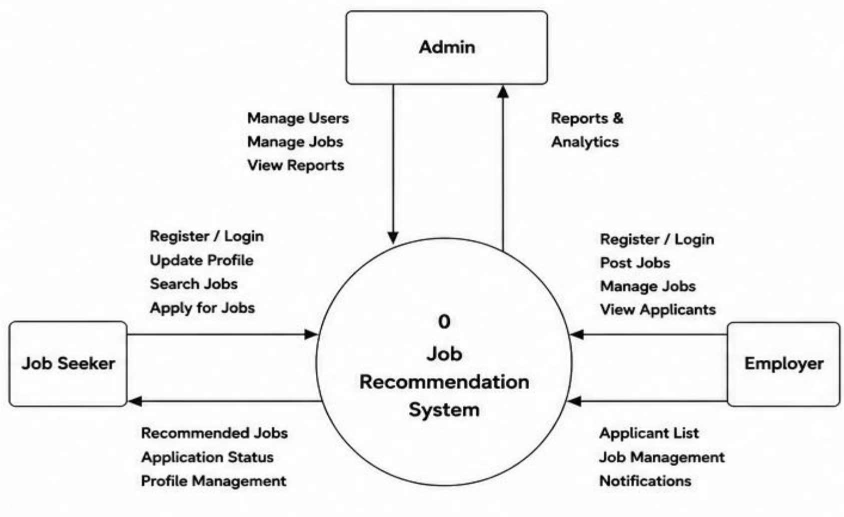

Figure 5: DFD Level 0 Diagram

DFD Level 1

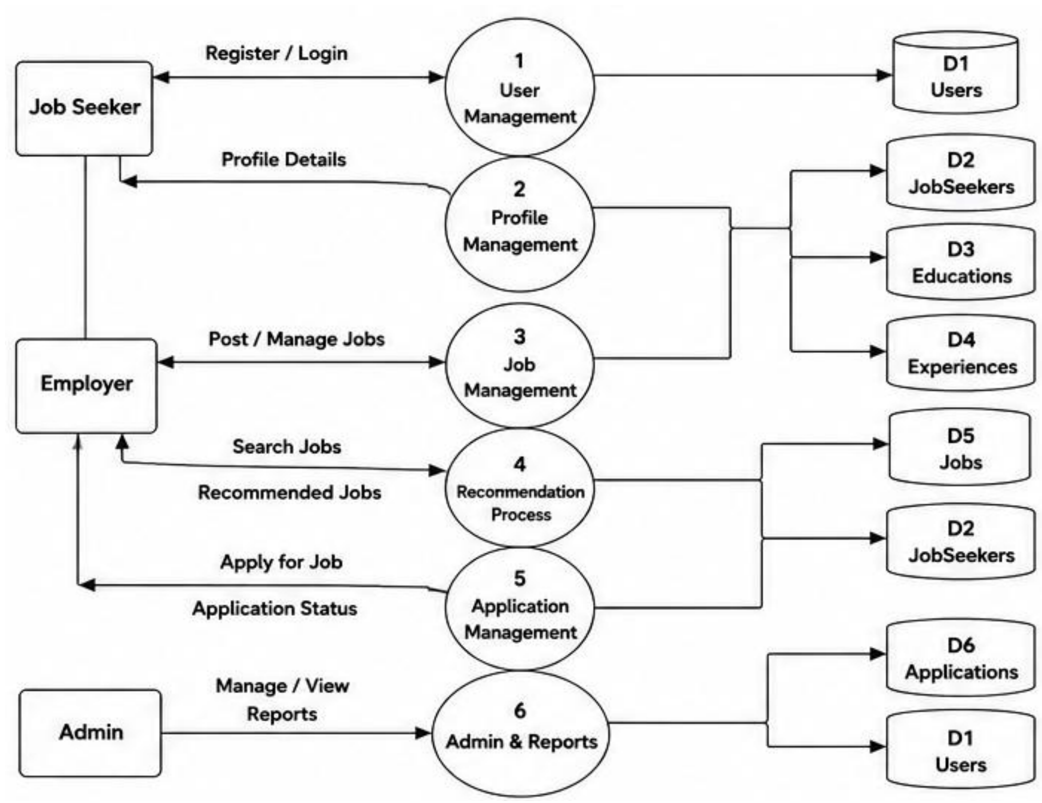

Figure 6: DFD Level 1 Diagram

DFD Level 2

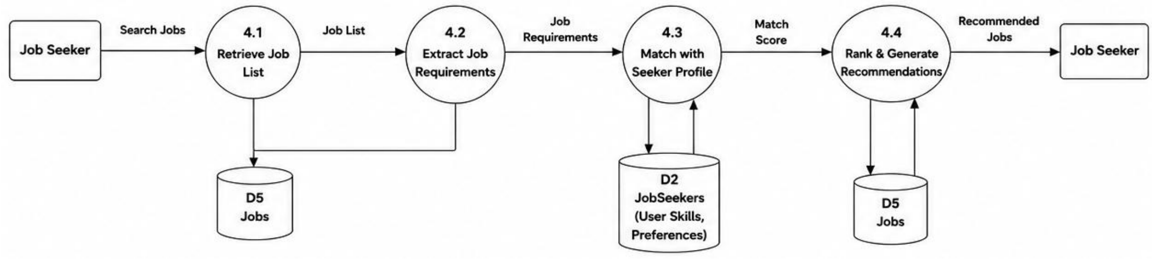

Figure 7: DFD Level 2 Diagram

Chapter 4 : System Design

4.1 System Design

System design is the process of defining the architecture, components, database structure, and user interface of a system based on the requirements identified during the analysis phase. It provides a blueprint for implementing the proposed system and ensures that all system components work together efficiently.

The proposed Job Recommendation System is designed as a three-tier web application consisting of a presentation layer, an application layer, and a database layer. The presentation layer provides an interactive user interface for job seekers, employers, and administrators. The application layer processes user requests, implements business logic, and generates personalized job recommendations. The database layer stores user information, job details, applications, skills, and other system data required for the operation of the application.

4.1.1 System Architecture

The proposed Job Recommendation System follows a three-tier architecture consisting of the Presentation Layer, Application Layer, and Database Layer. The Presentation Layer provides separate interfaces for Job Seekers, Employers, and Administrators using React.js. The Application Layer, developed with ASP.NET Core Web API, processes user requests, implements business logic, integrates the Google Gemini API for AI-based job description generation, and performs job recommendations. The Database Layer uses SQL Server to store user profiles, job postings, applications, and other system data.

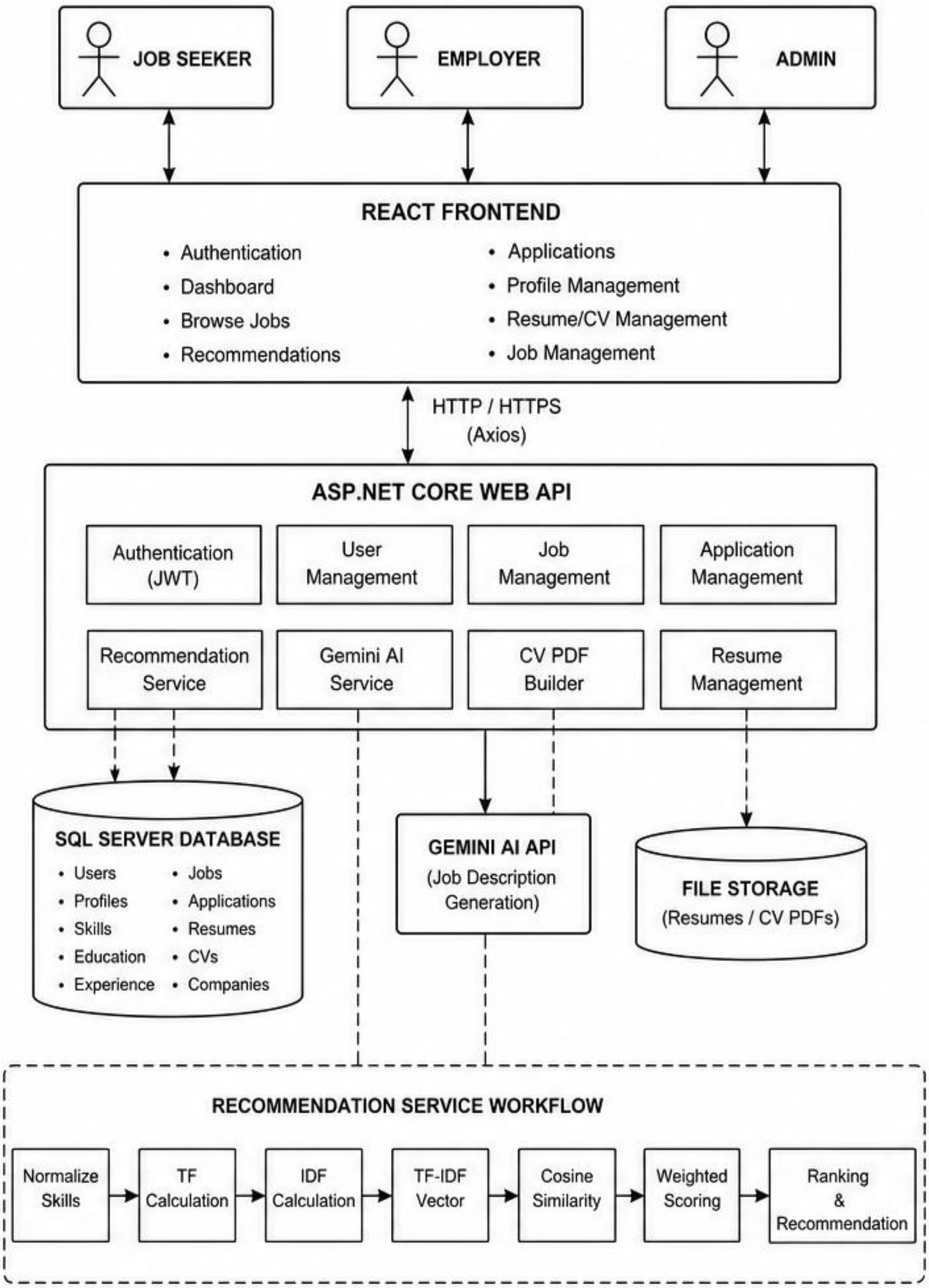

Figure 8: System Architecture Diagram

4.1.2 Relational Schema Design

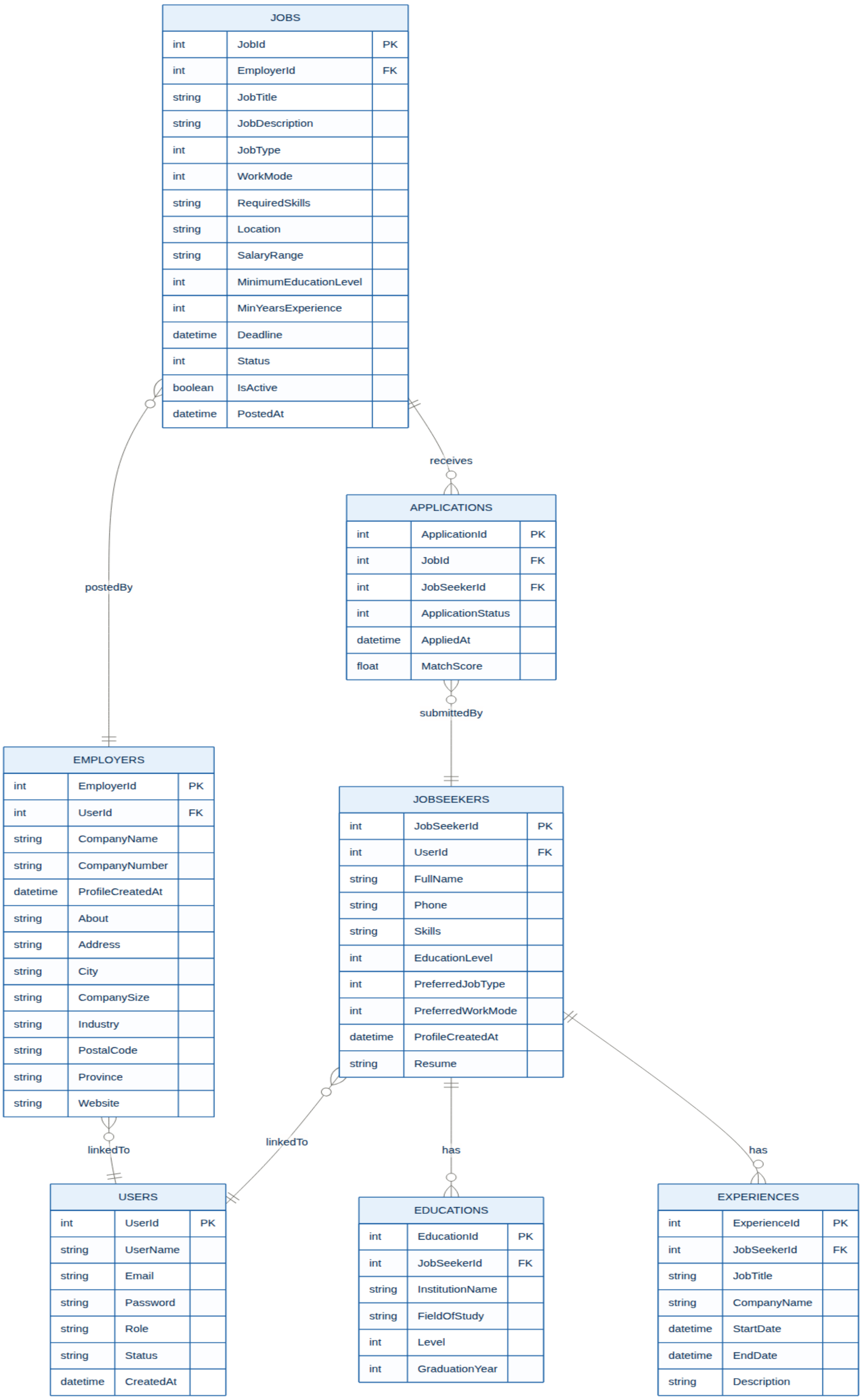

Figure 9: Relational Schema Design

4.1.3 Form Design

Form design defines the structure and layout of the input screens used in the system. The forms are designed to be simple, user-friendly, and responsive, allowing users to enter accurate information with minimal effort. Proper validation is applied to required fields to reduce input errors and improve data quality.

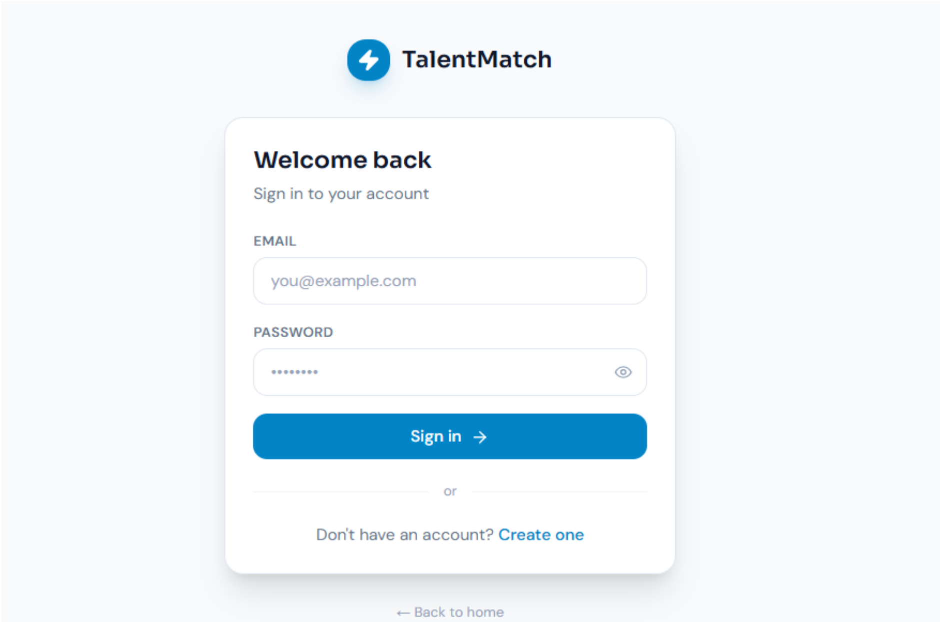

Figure 11: Login Page

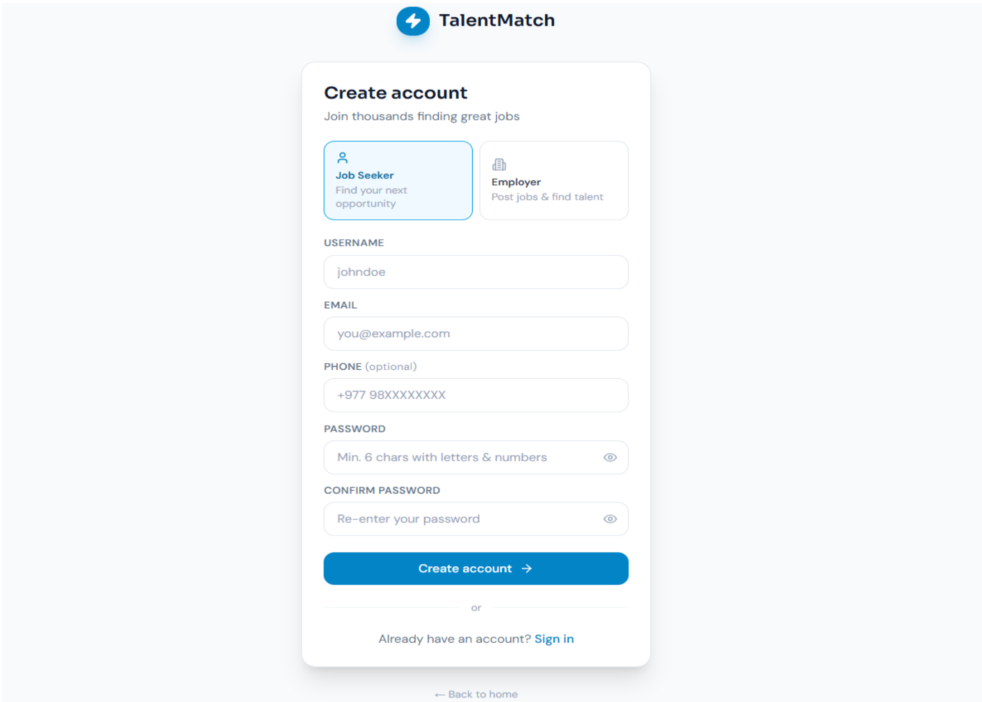

Figure 10: SignUp Page for Job Seeker

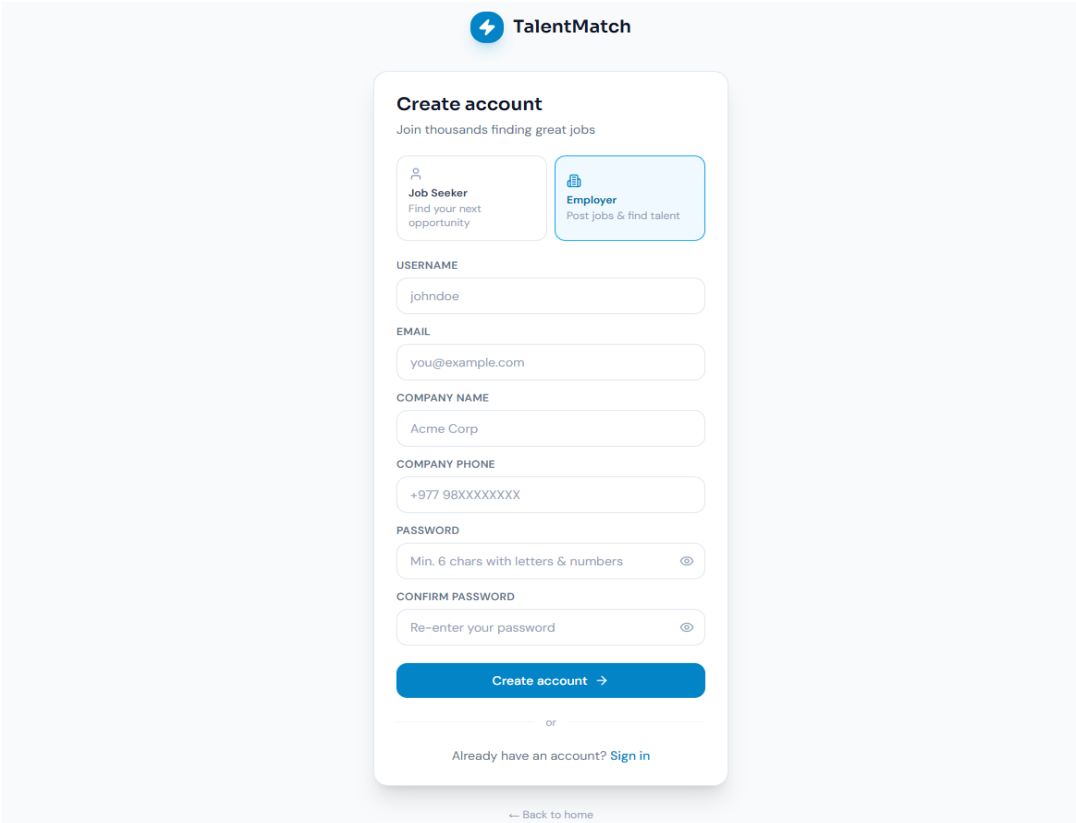

Figure 12: SignUp Page for Employer

4.1.4 User Interface Design

The user interface of the proposed Job Recommendation System is designed to be simple, intuitive and easy to use. The interfaces were created using React.js to ensure consistency and responsiveness across all devices. The figure below shows the interfaces for the Job Seeker, Employers and Administrators. The interfaces were designed to be user-friendly and increase interaction and readability. The interfaces were created using React.js and have a similar design to ensure there is consistency while using the application.

The major user interfaces of the system include:

- Login Page
- Registration Page
- Landing Page
- Job Seeker Dashboard
- Employer Dashboard
- Admin Dashboard
- Browse Jobs Page

- Job Recommendation Page
- Job Details Page
- Job Posting Page
- Candidate Page
- Manage Users Page

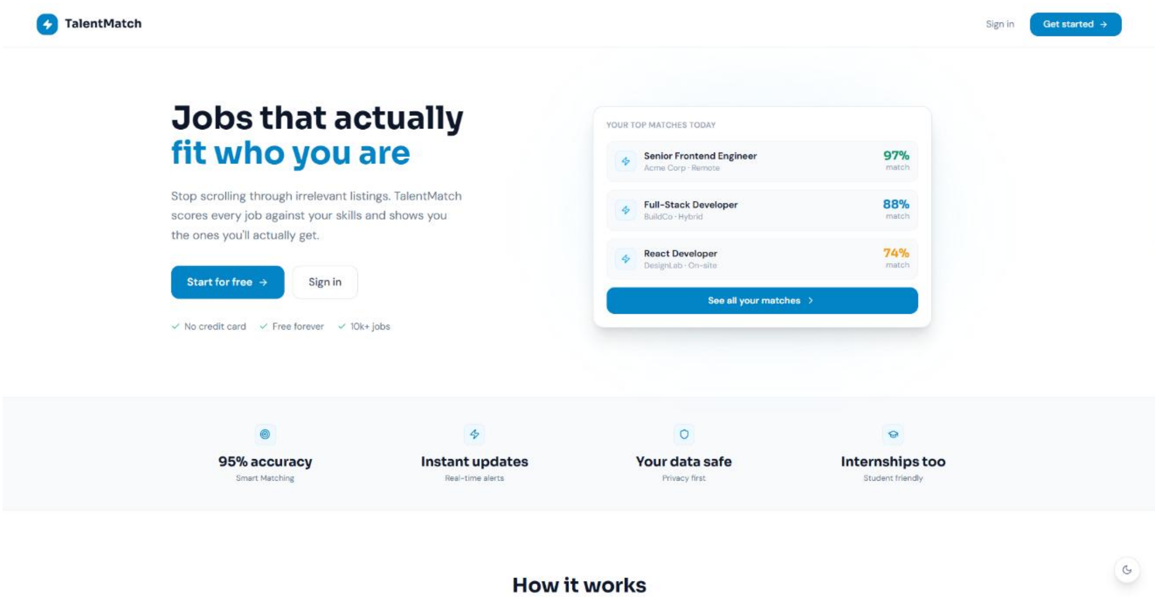

Figure 13: User Interface

4.1.5 AI-based Job Description Generation

The proposed system integrates the Google Gemini API to assist employers in creating job postings. When an employer enters a job title, the system sends the title to the Gemini API, which generates an appropriate job description, key responsibilities, and required skills. Employers can review and edit the generated content before publishing the job posting. This feature reduces manual effort, ensures consistency, and helps create comprehensive job advertisements.

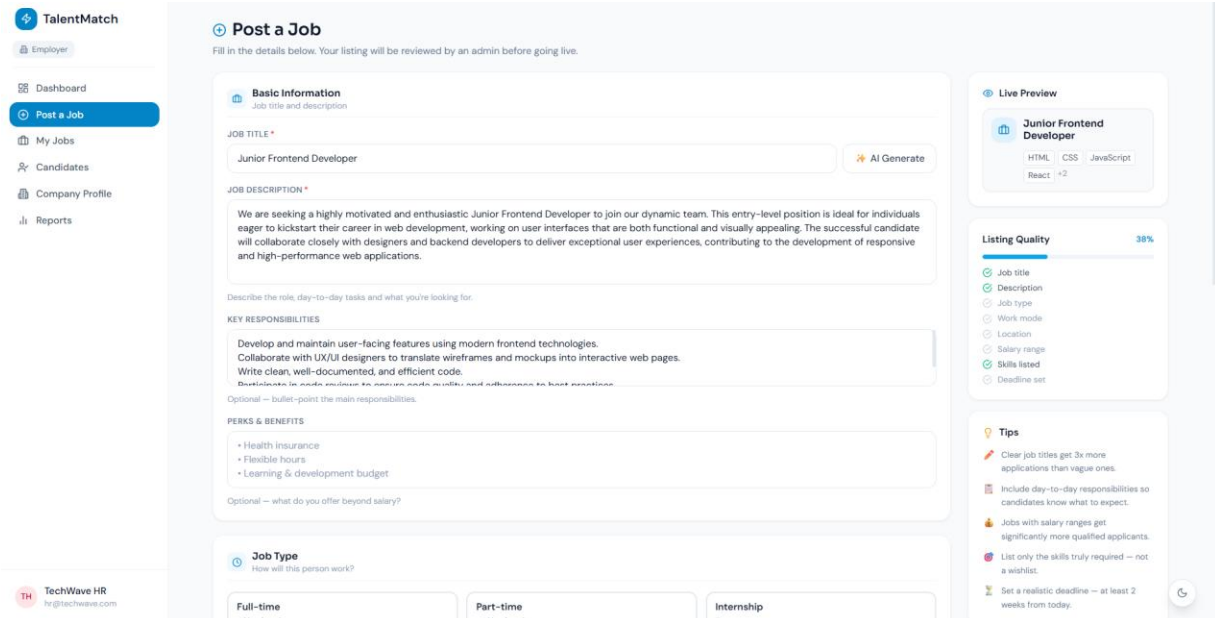

Figure 14: AI-based Job Description Generation Diagram

4.1.6 CV PDF Generation

The proposed system provides an automated CV generation feature that enables job seekers to create professional resumes directly from their profile information. The **CVPdfBuilder** module collects details such as personal information, education, work experience, projects, certifications, and skills, and formats them into a structured PDF document. Users can preview and download the generated CV without manually designing a resume.

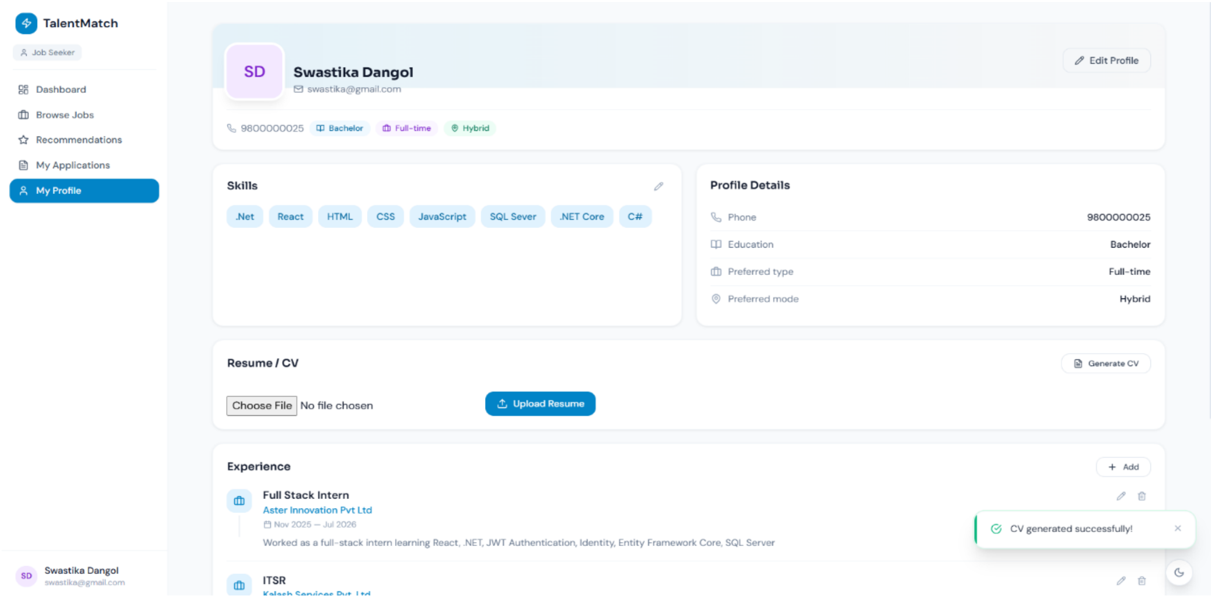

Figure 15: CV PDF Generation Diagram

Swastika Dangol

FullTime

9800000025 | swastika@gmail.com

SUMMARY

Full Stack Intern with 1+ year of experience, most recently at Aster Innovation Pvt Ltd. Strong technical foundation in .Net, React and HTML, with a track record of applying these skills to deliver practical results. Holds a Bachelor-level qualification and is committed to continuous learning and professional growth.

EXPERIENCE

Full Stack Intern — Nov 2025 - Jul 2026

Aster Innovation Pvt Ltd

- Worked as a full-stack intern learning React,
- NET, JWT Authentication, Identity, Entity Framework Core, SQL Server

ITSR — Jun 2022 - Feb 2023

Kalash Services Pvt. Ltd

- Provided customer support for an internet service provider, assiting customers with inquires, basic troubleshooting and resolving service-related isssues while ensuring a positive cutomer experience

EDUCATION

Bachelor in CSIT — 2026

TU

HighSchool in Science — 2022

NEB

SKILLS

- .Net
- HTML
- JavaScript
- .NET Core
- React
- CSS
- SQL Sever
- C#

Generated via TalentMatch

Figure 16: CV Generated by CV PDF Generator

4.2 Algorithm Details

The proposed Job Recommendation System employs a Hybrid Content-Based Job Recommendation Algorithm using TF-IDF and Cosine Similarity to generate personalized job recommendations. The recommendation process begins by normalizing the skills of both job seekers and job postings to ensure consistent comparison. Term Frequency (TF) and Inverse Document Frequency (IDF) are then calculated to determine the importance of each skill, and TF-IDF vectors are generated for both candidate profiles and job descriptions. The similarity between these vectors is computed using the Cosine Similarity algorithm to obtain the skill match score. To improve recommendation accuracy, the skill similarity score is combined with additional criteria, including experience, education level, job type, and work mode, using a weighted scoring approach. Finally, all jobs are ranked according to their final recommendation scores and presented to the job seeker in descending order, ensuring that the most relevant job opportunities appear first.

TF-IDF Algorithm

Step 1: Start.

Step 2: Retrieve the skills of the job seeker and the required skills from all approved job postings.

Step 3: Convert all skill names into lowercase.

Step 4: Split the skill string using commas (,), semicolons (;), or vertical bars (|).

Step 5: Remove leading and trailing spaces from each skill and store the normalized skills as a list.

Step 6: Count the occurrence of each normalized skill in the candidate profile and each job posting.

Step 7: Calculate the total number of skills in each candidate profile and job posting.

Step 8: Compute the Term Frequency (TF) using:

$$TF(t,d) = \frac{\text{Number of occurrences of skill } t}{\text{Total number of skills}}$$

Where:

- **t** = Skill
- **d** = Candidate profile or job description

Step 9: Count the number of job postings containing each skill.

Step 10: Compute the Inverse Document Frequency (IDF) using:

$$IDF(t) = \log\left(\frac{N}{1 + df(t)}\right)$$

Where:

- **N** = Total number of job postings
- **df(t)** = Number of job postings containing skill *t*

Step 11: Calculate the TF-IDF weight using:

$$TFIDF(t, d) = TF(t, d) \times IDF(t)$$

Step 12: Generate the TF-IDF vector for the candidate profile and each job posting.

Step 13: Store the TF-IDF vectors.

Step 14: Stop.

Cosine Similarity Algorithm

Step 1: Start.

Step 2: Retrieve the TF-IDF vectors of the candidate profile and each job posting.

Step 3: Compute the dot product of the candidate and job TF-IDF vectors.

Step 4: Compute the magnitude of each TF-IDF vector.

Step 5: Calculate the cosine similarity using:

$$Similarity(P, J) = \frac{P \cdot J}{\parallel P \parallel \times \parallel J \parallel}$$

Where:

- **P** = Job Seeker TF-IDF Vector
- **J** = Job TF-IDF Vector

Step 6: Convert the similarity value into a percentage.

$$SkillSimilarity = Similarity(P, J) \times 100$$

Step 7: Store the skill similarity score.

Step 8: Stop.

Weighted Scoring Algorithm

Step 1: Start.

Step 2: Retrieve the skill similarity score from the Cosine Similarity algorithm.

Step 3: Calculate the experience score.

- If Candidate Experience ≥ Required Experience, then:

$$ExperienceScore = 100$$

- Otherwise:

$$ExperienceScore = \frac{\text{Candidate Experience}}{\text{Required Experience}} \times 100$$

Step 4: Calculate the education score.

- Matching education level = **100**
- Otherwise = **0**

Step 5: Calculate the job type score.

- Matching job type = **100**
- Otherwise = **0**

Step 6: Calculate the work mode score.

- Matching work mode = **100**
- Otherwise = **0**

Step 7: Calculate the final recommendation score using:

$$FinalScore = (S \times 0.60) + (E \times 0.15) + (Ed \times 0.10) + (JT \times 0.10) + (WM \times 0.05)$$

Where:

- **S** = Skill Similarity Score
- **E** = Experience Score
- **Ed** = Education Score
- **JT** = Job Type Score
- **WM** = Work Mode Score

Step 8: Store the final recommendation score.

Step 9: Stop.

Job Ranking Algorithm

Step 1: Start.

Step 2: Retrieve all approved job postings and their final recommendation scores.

Step 3: Apply the selected filters (e.g., job type, work mode, or location), if any.

Step 4: Sort the job postings in descending order of their final recommendation scores.

Step 5: Assign a rank to each job based on its final recommendation score.

Step 6: Display the ranked job postings as personalized recommendations to the job seeker.

Step 7: Stop.

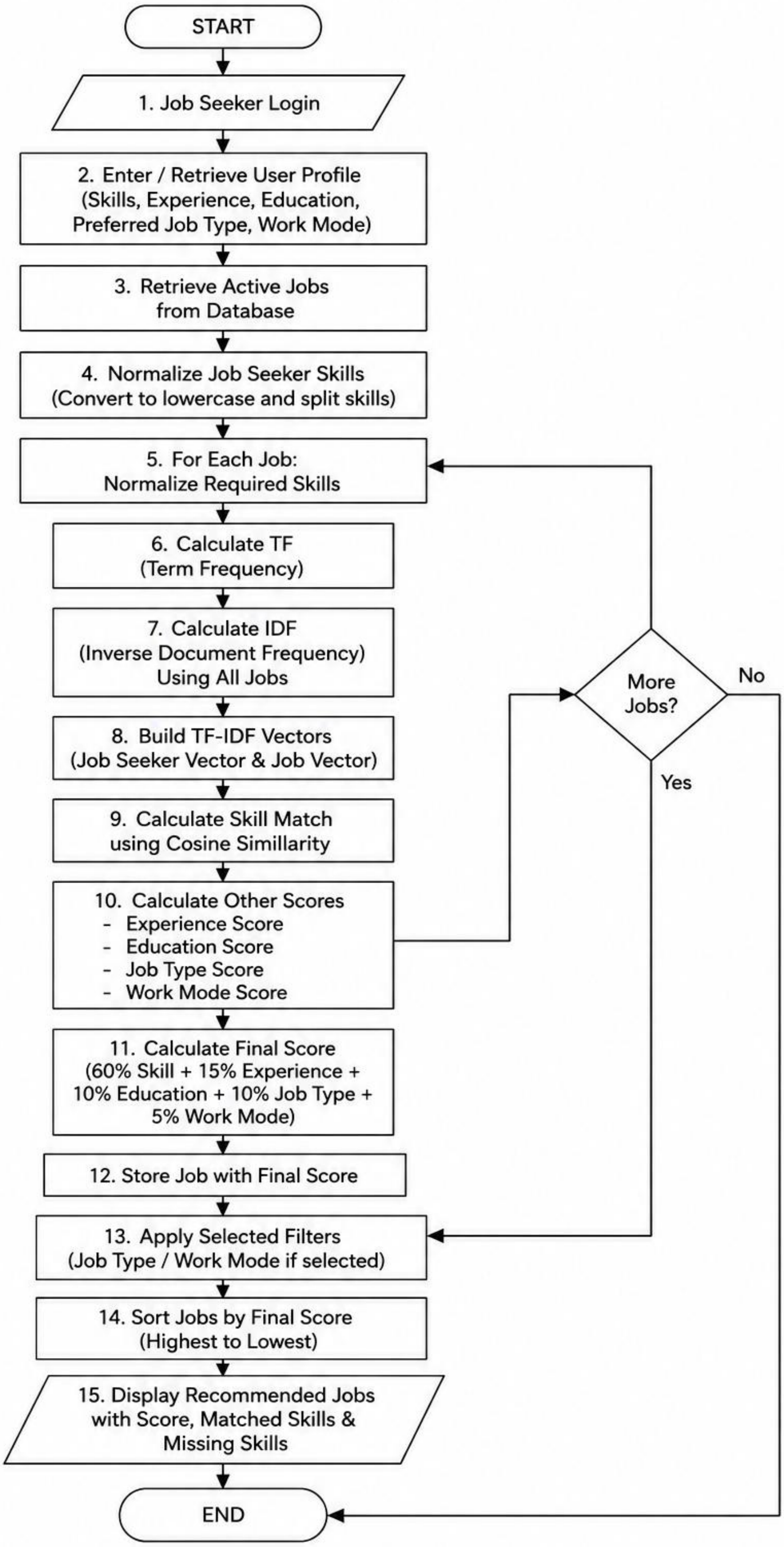

Figure 17: Flowchart of Algorithm

Chapter 5 : Implementation and Testing

5.1 Implementation

The implementation phase of the Job Recommendation System transforms the system design into a fully functional web-based application. This phase involves developing the frontend and backend components, designing the database, integrating authentication and authorization, implementing the recommendation module, and testing the complete system. The frontend is developed using React.js, while the backend is implemented using ASP.NET Core Web API. Microsoft SQL Server is used for database management, and Entity Framework Core provides database interaction through Object Relational Mapping (ORM). Secure user authentication and authorization are implemented using JSON Web Token (JWT). The recommendation module integrates TF-IDF, Cosine Similarity, and Weighted Scoring techniques to generate personalized job recommendations for job seekers based on their skills, qualifications, and preferences.

5.1.1 Tools Used

Design and Development Tools

Table 1 : Design and Development Tools

Category	**Tool/Platform**	**Purpose**
Programming Language	C#, JavaScript	Backend and frontend development
Frontend Framework	React.js	User interface development
Styling	Tailwind CSS	Reponsive and modern user interface
Backend Framework	ASP.NET Core Web API	Business Logic and RESTful API development
Database	Microsoft SQL Server	Data Storage and management
ORM	Entity Framework Core	Database access and object mapping
Authentication	JSON Web Toke(JWT)	User authentication and authorization

HTTP Client	Axios	Communication between frontend and backend
Development Tools	Visual Studio 2022, Visual Studio Code	Application Development
API Testing	Swagger	API testing and documentation
Version Control	Git and GitHub	Source Code Management
Diagram Tool	Draw.io	Use Case, DFD and ER diagrams

5.1.2 Implementation Details of Modules

Authentication Module

Table 2: Implementation Details of Authentication Modules

EndPoint	**Method**	**Description**
/api/auth/register	POST	Register a new user
/api/auth/login	POST	Authenticate user and generate JWT token

Job Seeker Module

Table 3: Implementation Details of Job Seeker Module

EndPoint	**Method**	**Description**
/api/JobSeeker/profile/{id}	GET, PUT	View and update job seeker profile
/api/JobSeeker/profile/{id}/upload-resume	POST	Upload resume
/api/JobSeeker/profile/{id}/delete-resume	DELETE	Delete uploaded resume
/api/JobSeeker/profile/{id}/education, /api/JobSeeker/education/{educationId}	POST, GET, PUT, DELETE	Add, view, update, and delete education details

/api/JobSeeker/profile/{id}/experience, /api/JobSeeker/experience/{experienceId}	POST, GET, PUT, DELETE	Add, view, update, and delete work experience
/api/JobSeeker/jobs/{id}	GET	View job details
/api/JobSeeker/recommendations/{id}	GET	Retrieve personalized job recommendations
/api/JobSeeker/apply	POST	Apply for a job
/api/JobSeeker/applications/{id}	GET	View submitted job applications
/api/JobSeeker/profile/{id}/generate-cv	GET	Generate CV in PDF format

Employer Module

Table 4: Implementation Details of Employer Module

EndPoint	**Method**	**Description**
/api/Employer/profile/{id}	GET, PUT	View and update employer profile
/api/Employer/postjobs/{id}	POST	Create a new job posting
/api/Employer/myjobs/{id}	GET	View posted jobs
/api/Employer/jobs/{jobId}	PUT, DELETE	Update and delete job postings
/api/Employer/jobs/{jobId}/applicants	GET	View applicants for a job
/api/Employer/applications/{applicationId}/status	PUT	Update application status
/api/Employer/reports/{employerId}	GET	View employer reports

Admin Module

Table 5: Implementation Details of Admin Module

EndPoint	**Method**	**Description**
/api/Admin/stats	GET	View dashboard statistics

/api/Admin/users	GET	View all users
/api/Admin/users/jobseekers	GET	View all job seekers
/api/Admin/users/employers	GET	View all employers
/api/Admin/users/{userId}	GET, DELETE	View and delete a user
/api/Admin/users/{userId}/toggle	PUT	Activate or deactivate a user account
/api/Admin/jobs	GET	View all job postings
/api/Admin/jobs/{jobId}	GET, DELETE	View and delete a job posting
/api/Admin/jobs/{jobId}/approve	PUT	Approve a job posting
/api/Admin/jobs/{jobId}/reject	PUT	Reject a job posting
/api/Admin/applications	GET	View all job applications

AI Module

Table 6: Implementation Details of AI Module

EndPoint	**Method**	**Description**
/api/AI/generate-job	POST	Generate AI-powered job description using Google Gemini

5.2 Testing

Testing was performed to ensure that each module in the Job Recommendation System works properly according to the requirements. Modules were tested through unit testing, and the entire system was subjected to systems testing in order to ensure the proper integration of all components, including authentication, job management, recommendation, and application.

5.2.1 Test Cases for Unit Testing

Unit Testing section shows that the proposed job recommendation system was initially tested to ensure that each module works before being integrated into the entire system. In the proposed system, several modules have been taken into consideration for testing based on their functionalities, such as Authentication, Job Seeker, Employer, Recommendation, AI-based Job Description Generation, CV Generation, and Admin. All modules have been tested separately by considering various test cases, and results demonstrate that each module performed well and achieved the expected functions.

Unit testing analysis results

Table 7: Test Cases for Unit Testing

Test ID	Module	Test Case	Input	Expected Outcome	Result
UT-01	Authentication	Register with real data	Valid name, email, password and role	User account is created successfully	Passed
UT-02	Authentication	Register with existing email	Existing email address	Registration is rejected with an error message	Passed
UT-03	Authentication	Login with valid credentials	Registered email and correct password	User is authenticated and JWT token is generated	Passed
UT-04	Authentication	Login with invalid credentials	Registered email and incorrect password	Authentication fails and an error messages is displayed	Passed
UT-05	Job Seeker	Update Profile	Modify profile information	Profile information is updated successfully	Passed
UT-06	Job Seeker	Add education	Valid education details	Education record is added successfully	Passed
UT-07	Job Seeker	Add work experience	Valid experience details	Experience record is added successfully	Passed

UT-08	Job Seeker	Upload resume	Upload a valid PDF resume	Resume is uploaded successfully	Passed
UT-09	Job Seeker	Generate CV	Click Generate CV	CV is generated successfully in PDF format	Passed
UT-10	Employer	Create job posting	Enter valid job details	Job posting is created successfully	Passed
UT-11	Employer	Generate AI Job description	Enter job title and click AI Generate	AI-generated job description is displayed	Passed
UT-12	Employer	Update job posting	Modify job details	Job posting is updated succesfully	Passed
UT-13	Employer	View applicants	Select a posted job	Applicant list is displayed successfully	Passed
UT-14	Recommendation	Generate job recommendations	Request recommendations	Personalized job recommendations are displayed	Passed
UI-15	Recommendation	Calculate match score	Compare candidate profile with a job	Match score is calculated successfully	Passed
UT-16	Admin	Approve job posting	Approve a pending job	Job status changed to Approved	Passed

UT-17	Admin	Reject job posting	Reject a pending job	Job status changes to Rejected	Passed
UT-18	Admin	Manage users	View and update user status	User information is managed successfully	Passed

5.2.2 Test Cases for System Testing

System testing was performed to ensure that all modules of the proposed Job Recommendation System were integrated correctly and operated as designed. The scope of the test included such features as user authentication, job posting, job recommendation, AI-based job description generation, application, CV generation, and user management. The test results were positive, as the system was confirmed to support all the required functions accordingly.

System testing analysis results

Table 8: Test Cases for System Testing

Test ID	Test Scenario	Input	Expected Outcome	Status
ST-01	User Registration	Register a new account	User account is created successfully	Passed
ST-02	User Login	Login with valid credentials	User is redirected to the appropriate dashboard	Passed
ST-03	Job Posting	Employers creates a job posting	Job is submitted for admin approval	Passed
ST-04	Job Approved	Admin approves a job posting	Approved job becomes visible to job seekers	Passed
ST-05	Job Recommendation	Job Seeker requests recommendations	Personalized job recommendations are displayed	Passed

ST-06	AI Job Description Generation	Employer generates a job description	AI-generated job description is displayed	Passed
ST-07	Job Application	Job Seeker applies for a job	Application is submitted successfully	Passed
ST-08	Applicant Management	Employer views applicants	Applicant list is displayed successfully	Passed
ST-09	CV Generation	Job seekers generate a CV	CV is generated successfully in PDF format	Passed
ST-10	User Logout	User logs out	User session is terminated successfully	Passed

5.3 Result Analysis

The created Job Recommendation System was successfully designed and tested to ensure the proper functioning of its functions. The system works by recommending jobs to job seekers based on their skills using the TF-IDF and Cosine Similarity algorithms and a weighted approach that involves the level of experience, the field of study, job type, and mode of work, among others. The recommended job opportunities were relevant to the users' preferences and improved the accuracy of the initial matches. Adding the Gemini API into the system allowed employers to generate job descriptions using the AI tool, thus saving time and effort. The system also had a feature that enabled job seekers to generate CV PDFs from their profiles. The tests showed that all the functions, including authentication, jobs, recommendations, Gemini API for job descriptions, CV generation, and admin, were working correctly and met the required standards. Thus, the proposed solution can be considered successful as it addressed the research problem of finding relevant internship and job opportunities appropriately.

Chapter 6 : Conclusion and Future Recommendation

6.1 Conclusion

The Job Recommendation System was successfully designed and developed to meet the needs of establishing an effective platform for connecting job seekers and employers through a personal recommendation system. The system offers job seekers a convenient way to set up their profiles, upload their resumes, search for jobs, receive relevant recommendations, and apply for their desired positions. Meanwhile, the system allows employers to post available job positions, manage applications, and screen candidates according to match percentages. In addition, the system provides administrators with the tools to manage other users and maintain system performance.

The recommendation system employs a hybrid content-based approach that recommends relevant jobs by analyzing the candidate's profile, employing skill normalization and TF-IDF feature extraction and selection, Cosine Similarity implementation, applying a weighted scoring mechanism, and ranking the results according to job scores. Therefore, the developed recommendation system is expected to reduce the time spent by job seekers searching for relevant job opportunities as well as the time spent by employers screening candidates. Besides, the recommendation system proved to be functional, reliable, and efficient in meeting the needs of the platform.

The proposed system also includes additional features such as job description generation using AI, offered by Google Gemini API, and CV generation in PDF, thus making the system more convenient for both employers and job seekers. The features eliminate additional work for employers since they can generate job descriptions easily and quickly and rely on AI to screen job applicants. On the other hand, the CV generation in PDF allows job seekers to prepare a professional resume for application. Therefore, the system's proposed features are functional, reliable, and efficient.

6.2 Limitations

Although the proposed system fulfils its objectives, it has several limitations:

- The accuracy of job recommendations depends on the completeness and correctness of the profile information provided by job seekers and employers.

- AI-generated job descriptions may require manual review and editing before they are published.
- The system relies on an active internet connection to access the Google Gemini API for AI-based job description generation.
- The recommendation system uses a content-based approach and does not consider user behaviour or application history for generating recommendations.

6.3 Future Improvements

Although the proposed Job Recommendation System successfully meets its objectives, there is scope for further enhancement. Future improvements can focus on improving recommendation accuracy, expanding system functionality, and enhancing the overall user experience. The following enhancements are suggested:

- Enhance the recommendation algorithm by incorporating collaborative filtering and user behavior analysis to improve recommendation accuracy.
- Improve the AI-based job description generation feature to produce more detailed and context-aware job descriptions with minimal manual editing.
- Integrate real-time notifications for new job postings, application status updates, and personalized job recommendations.
- Provide multiple professional CV templates, allowing job seekers to choose from different resume designs and customize the appearance of their generated CVs.

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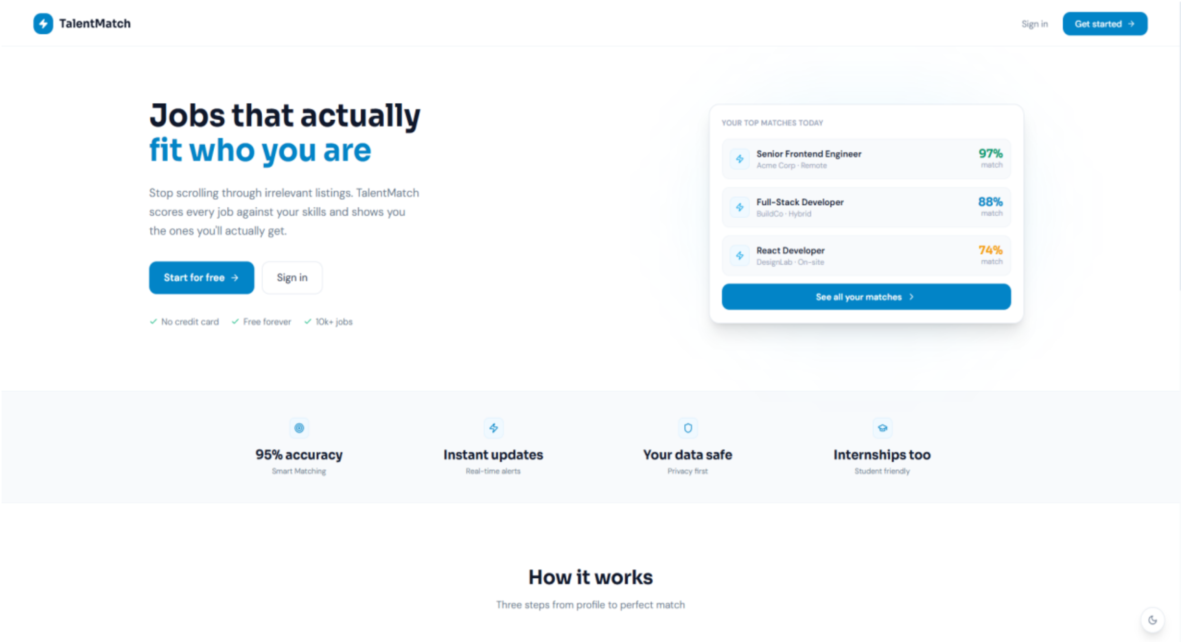
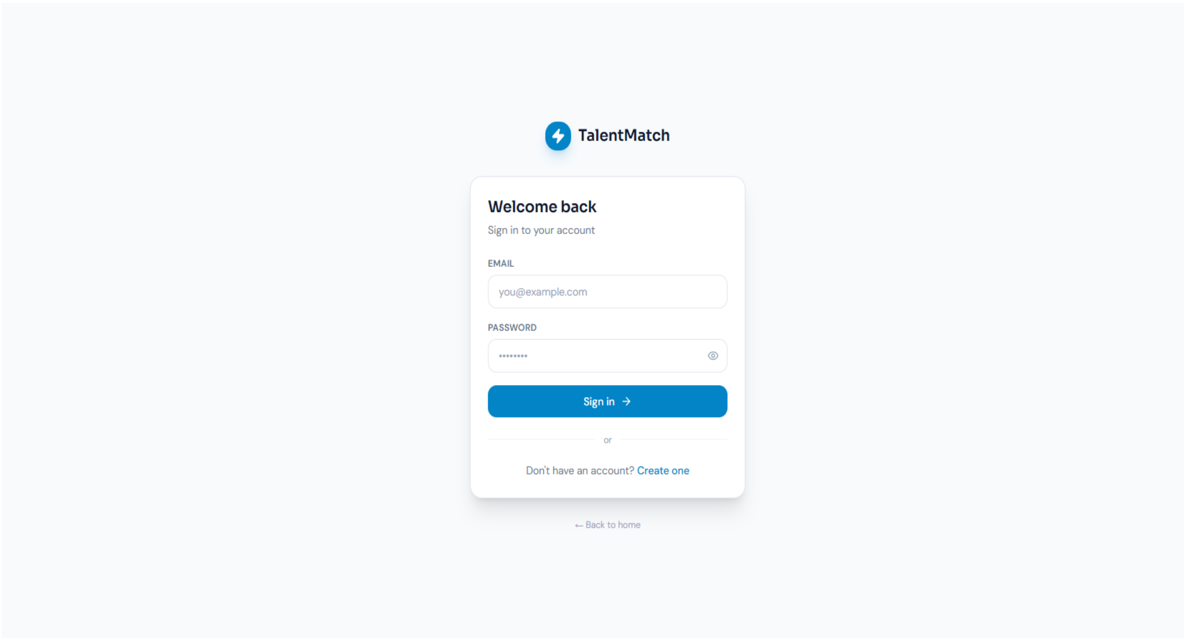
Appendix A : Interface Structure

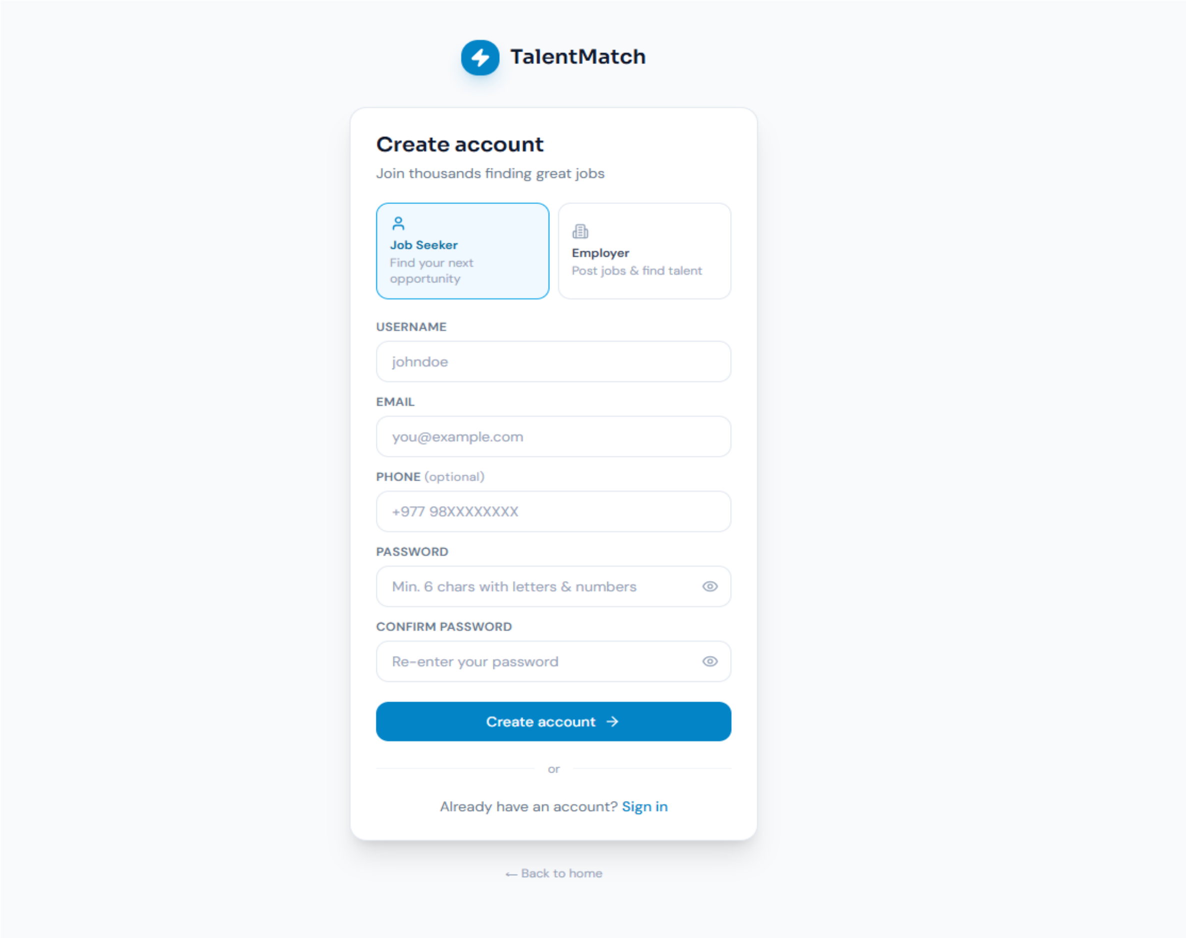
Landing Page



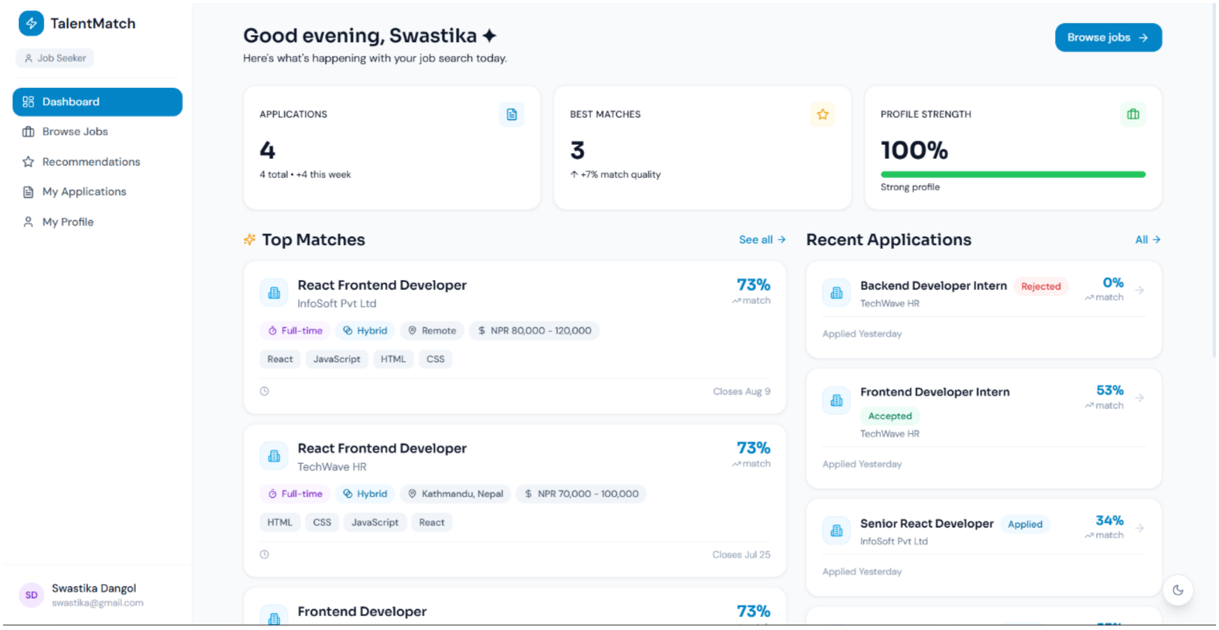
Login Page



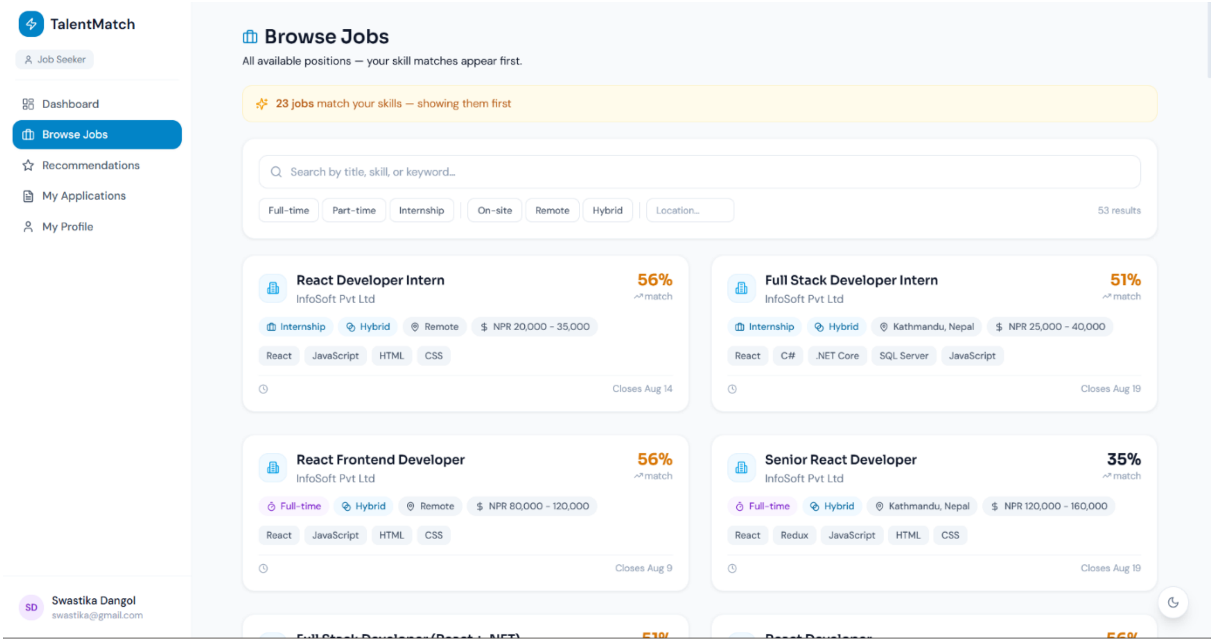
Register Page



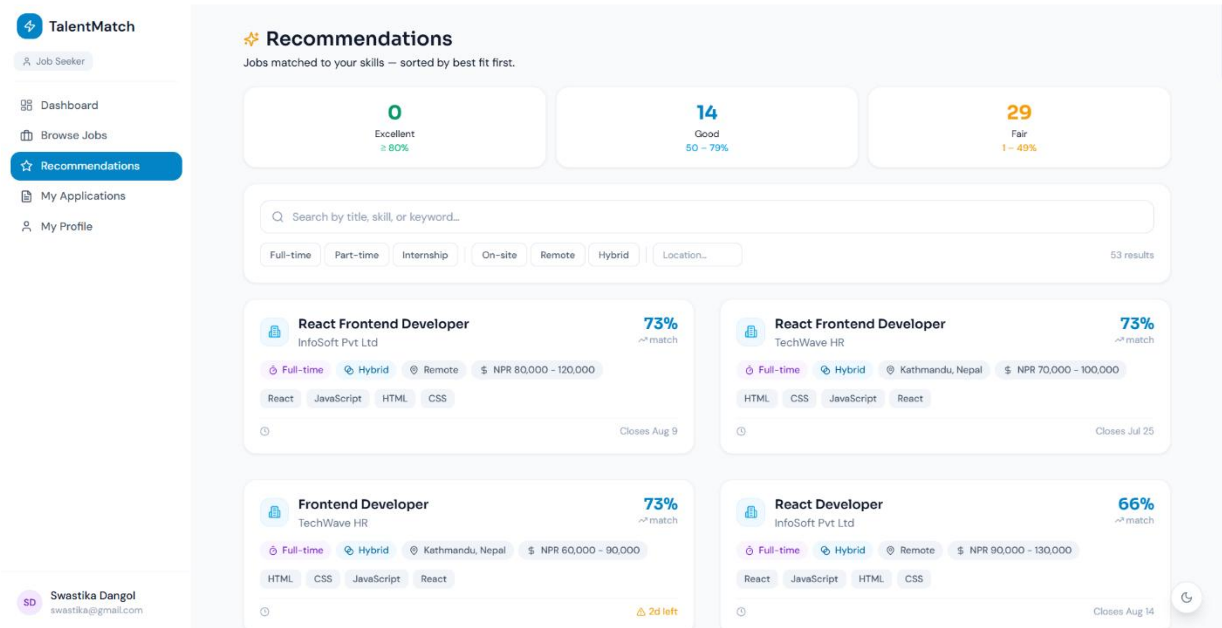
Job Seeker Dashboard



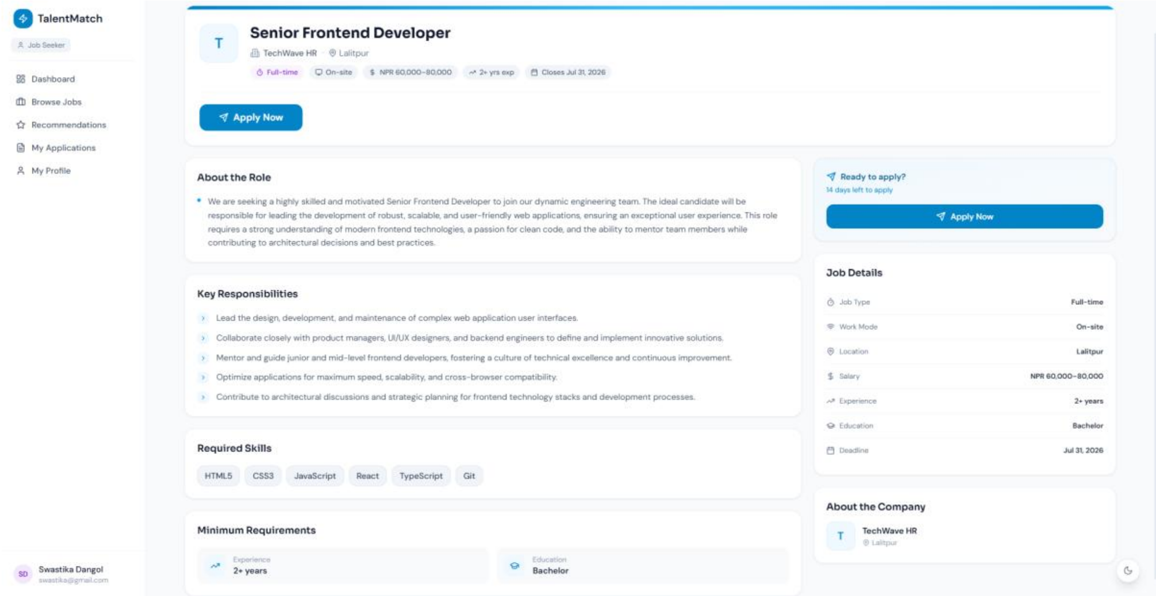
Browse Jobs Page



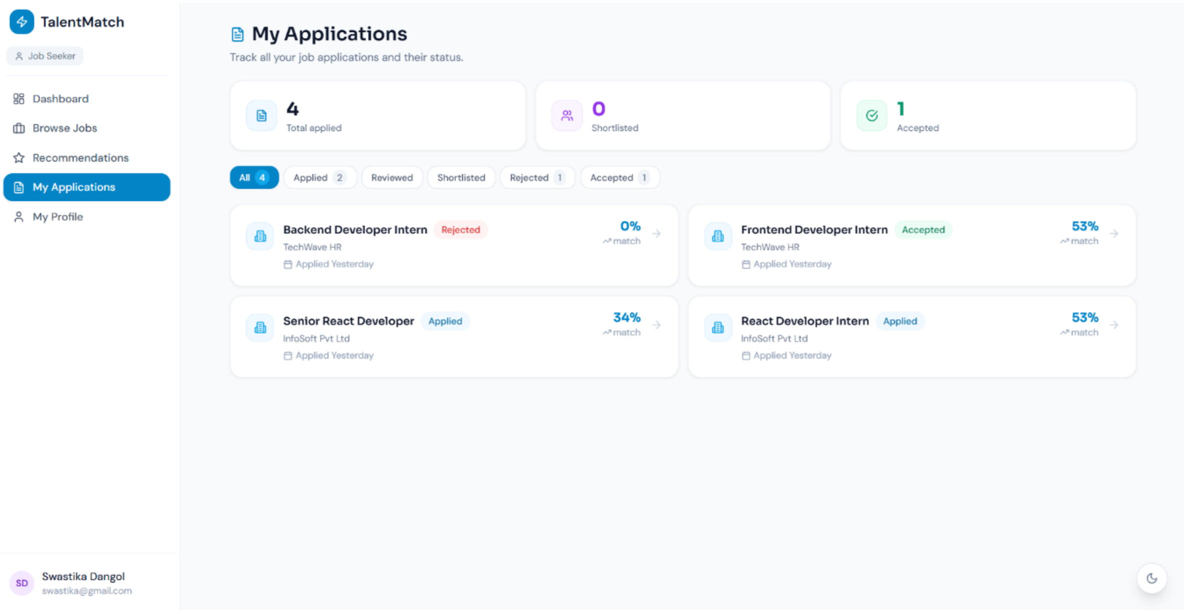
Recommendation Page



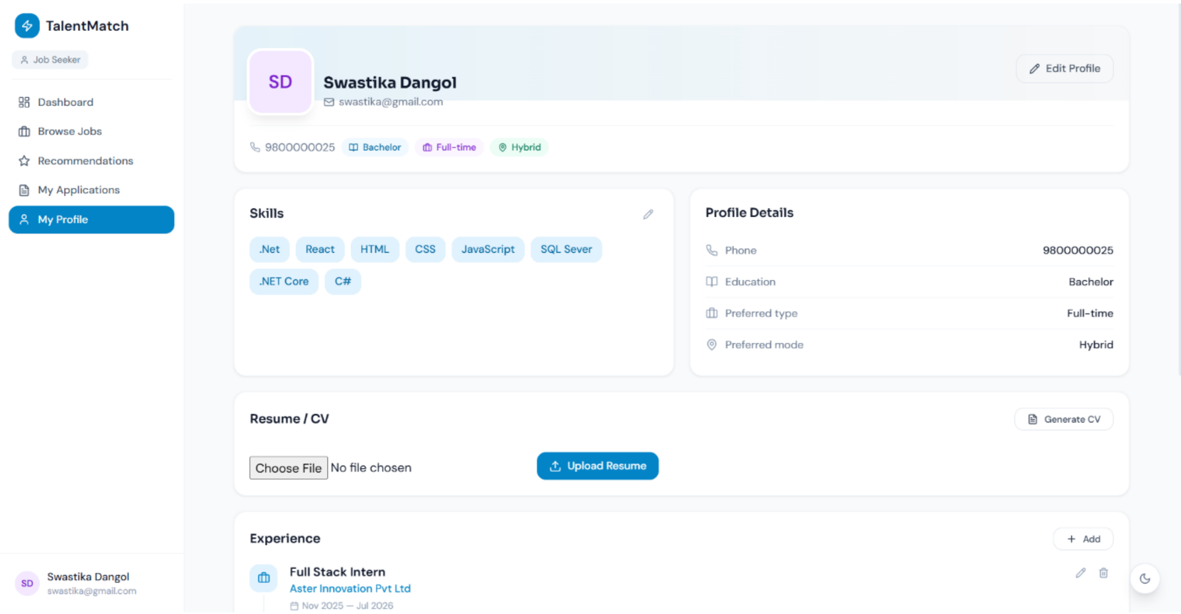
Job Details Page



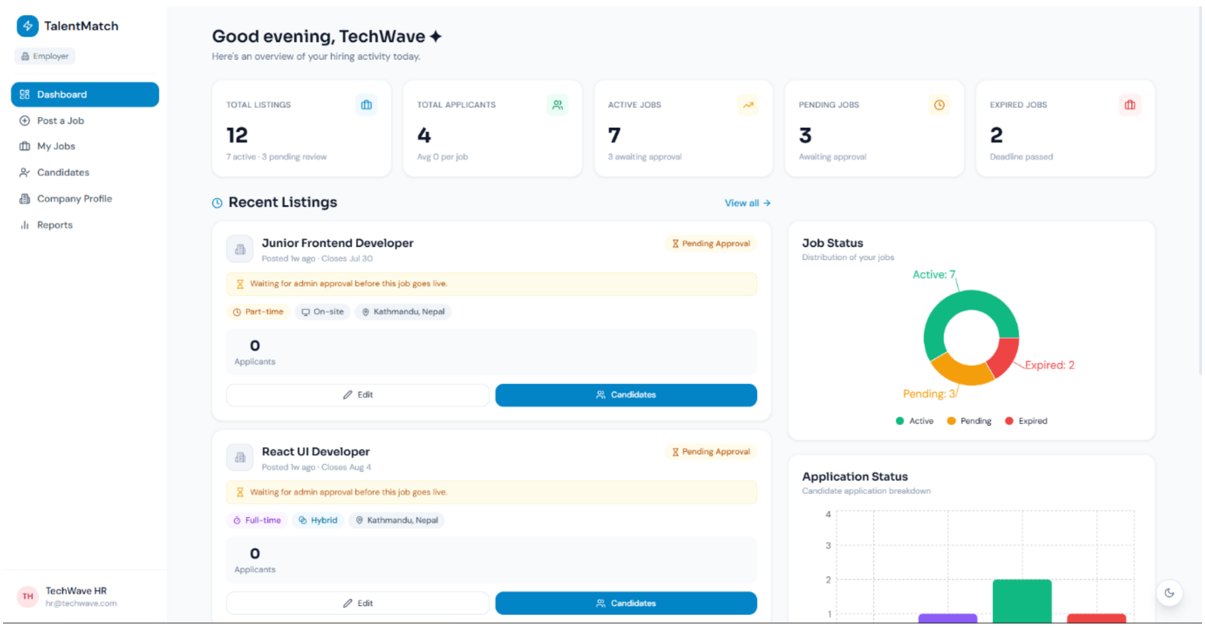
My Application Page



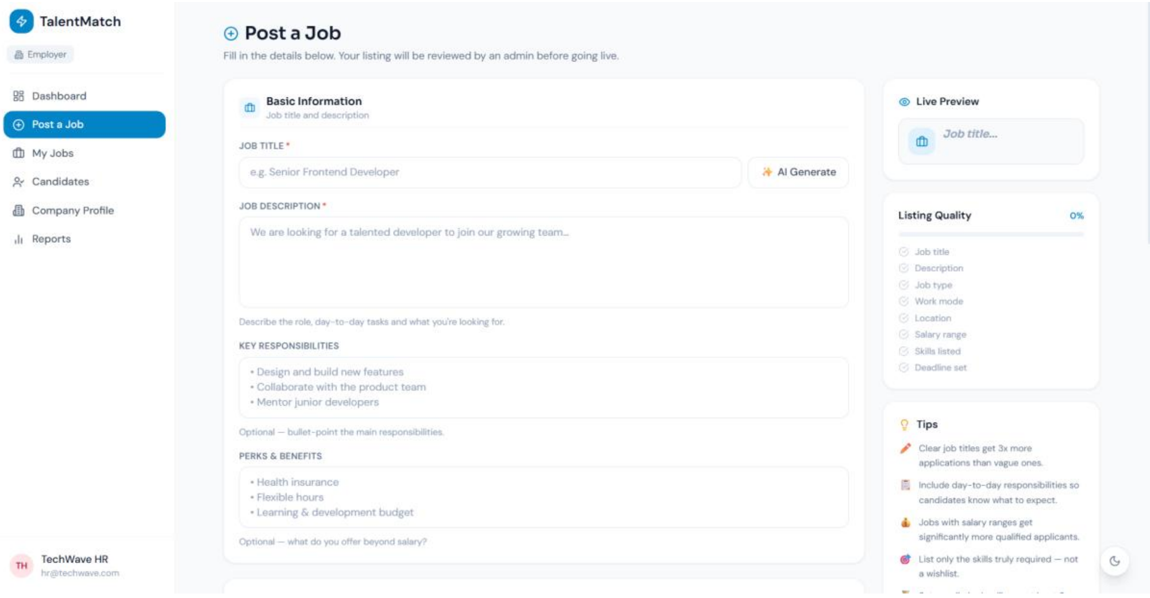
Job Seeker Profile Page



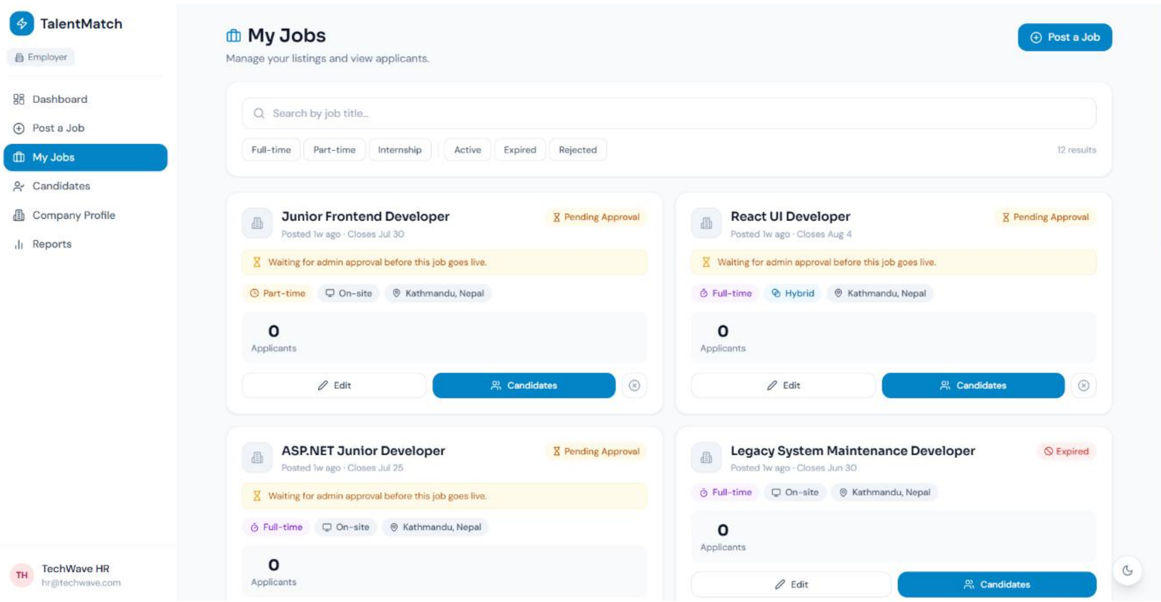
Employer Dashboard



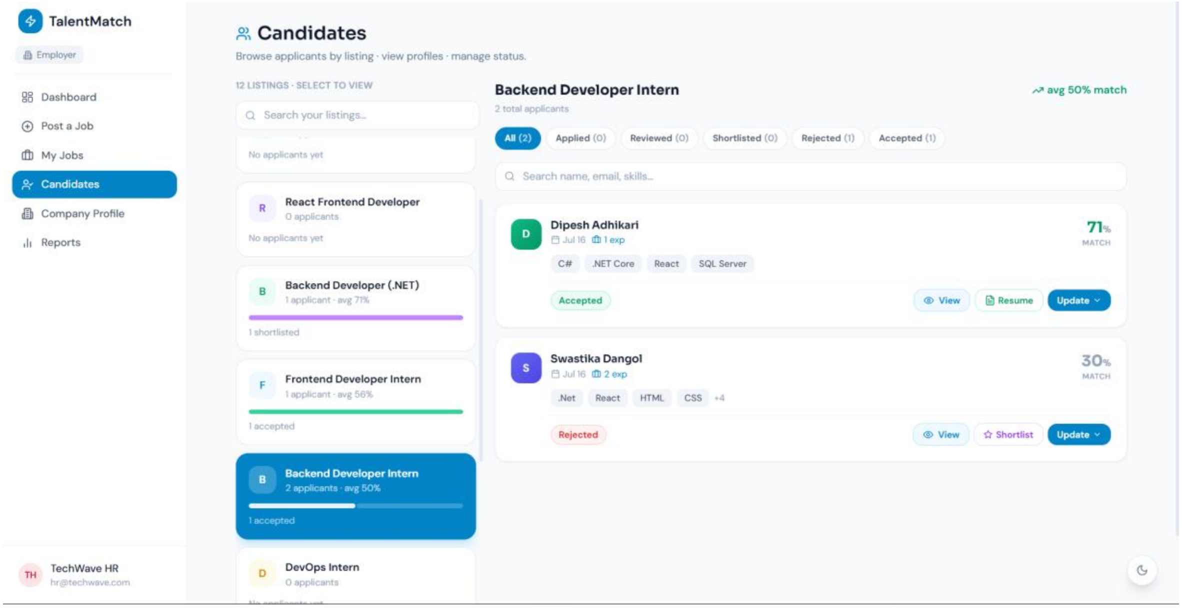
Post a Job Page



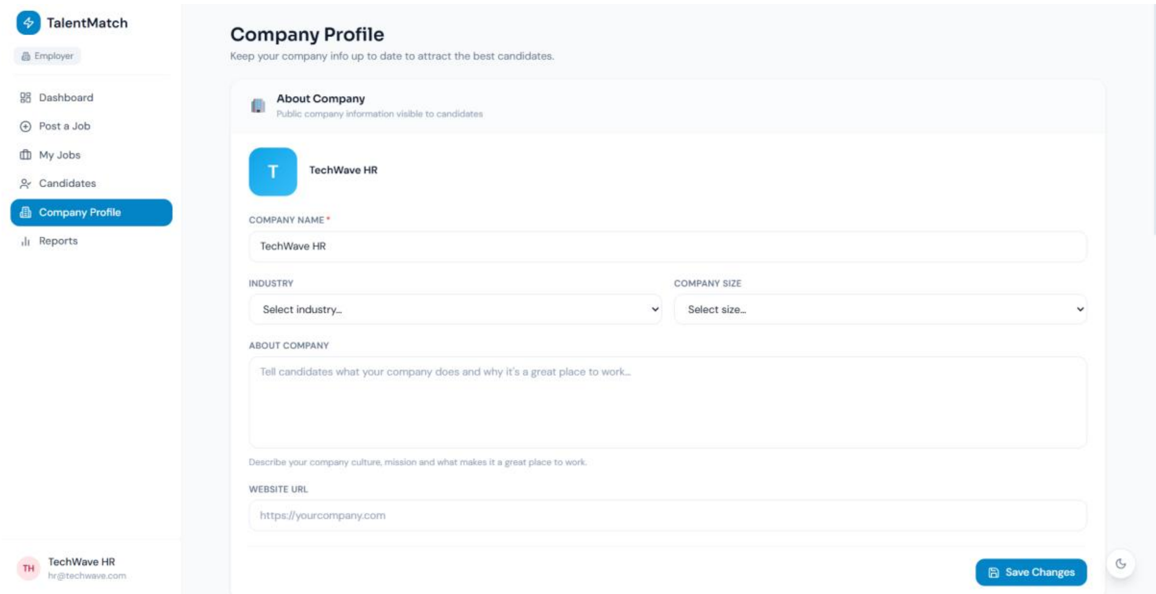
My Jobs Page



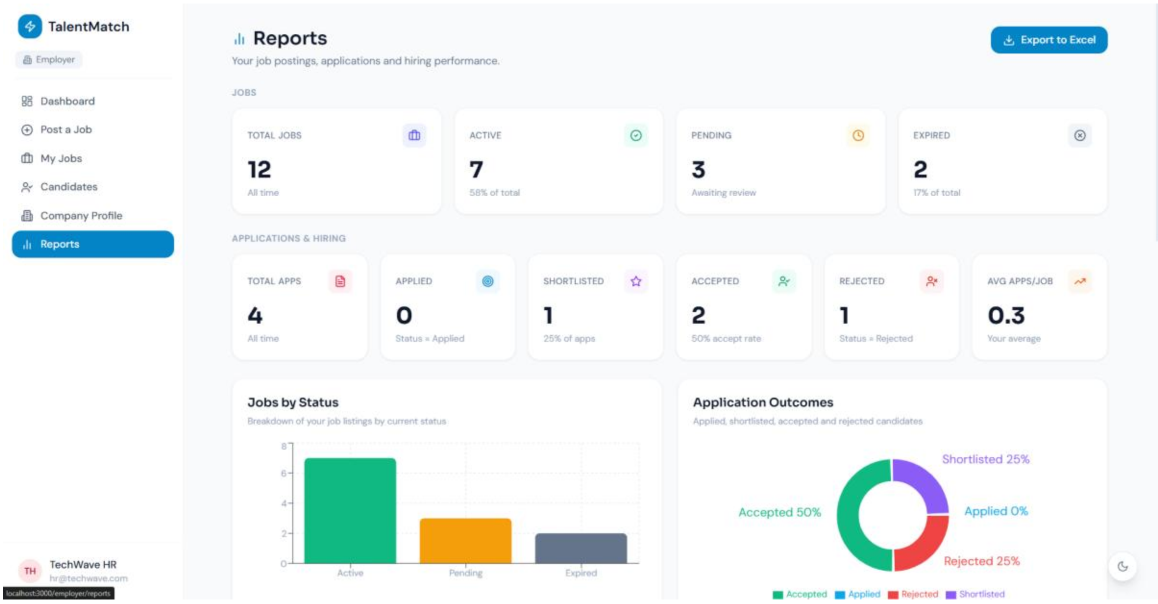
Candidates Page



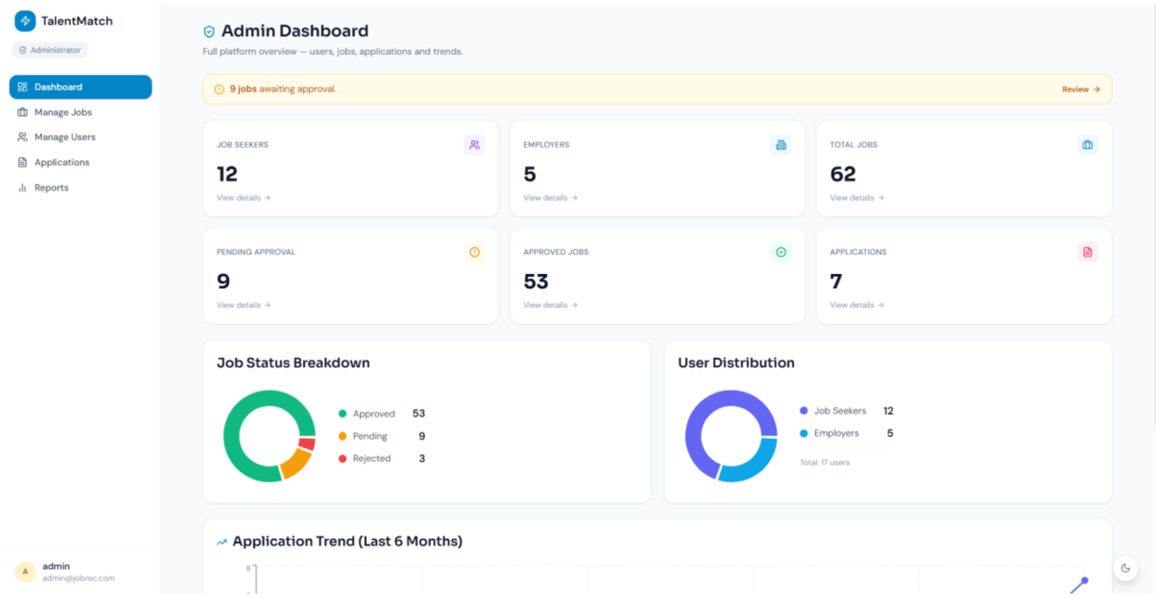
Company Profile Page



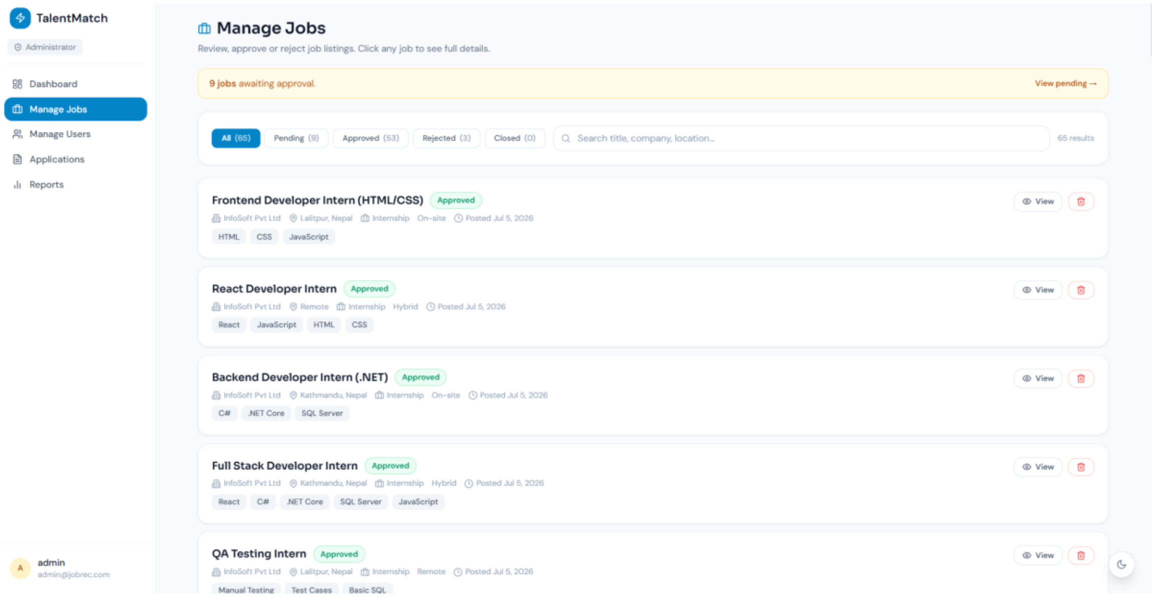
Employer Reports Page



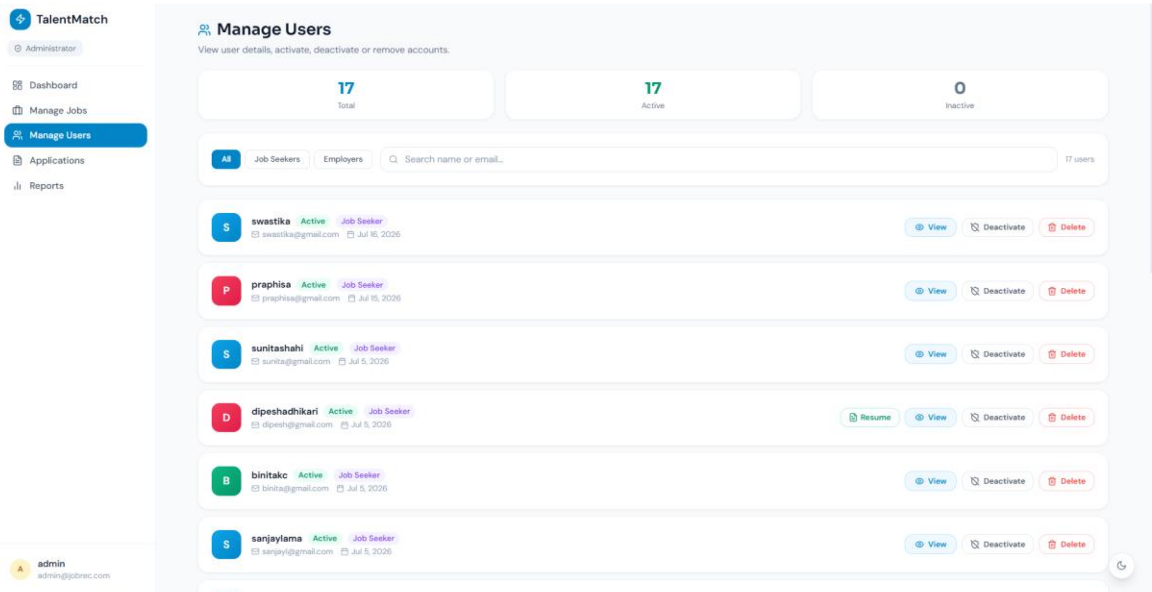
Admin Dashboard Page



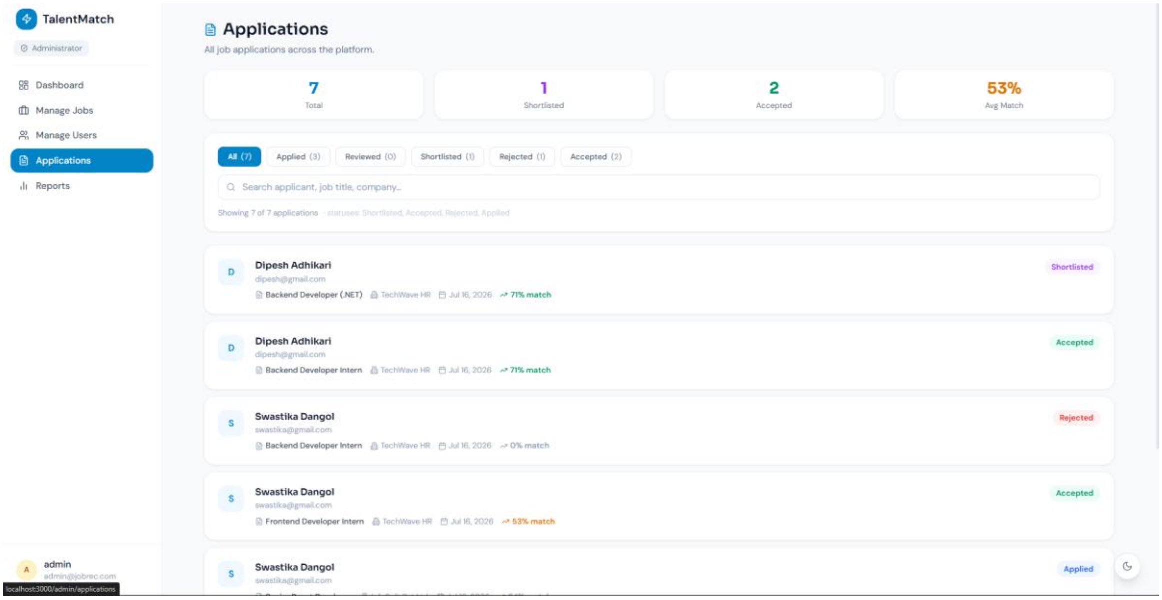
Manage Jobs Page



Manage Users Page



Applications Page



Admin Reports Page